

# Complete Problem Index — by Template Category

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All **737 problems** from `LeetCode_Complete_Reference.tex` , assigned to their template category. ★ marks problems with a worked template/solution already in this file's category sections above; unmarked problems are catalogued here for coverage. Source category shown when it differs (e.g. Trees & Backtracking → DFS; Heap → Stack/Queue).

## Two Pointers (93 problems, 93 templated)

| #   | ★ | Name                                   | Complexity      | Description                                                                                                                                                       |
|-----|---|----------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | ★ | Two Sum                                | $O(n) / O(1)$   | Given an integer array and a target sum, return the indices of two numbers that add up to target. Exactly one solution exists; cannot use the same element twice. |
| 11  | ★ | Container With Most Water              | $O(n) / O(1)$   | Given n vertical lines at positions 1..n with heights, find two lines that together with the x-axis form a container holding the most water.                      |
| 15  | ★ | 3Sum                                   | $O(n^2) / O(n)$ | Find all unique triplets [a,b,c] in an integer array such that $a+b+c=0$ . No duplicate triplets.                                                                 |
| 16  | ★ | 3Sum Closest                           |                 | Find three integers closest to a given target sum. Return the sum of those three integers.                                                                        |
| 18  | ★ | 4Sum                                   | $O(n^3) / O(n)$ | Find all unique quadruplets in an array that sum to a given target. No duplicates.                                                                                |
| 26  | ★ | Remove Duplicates from Sorted Array    | $O(n) / O(1)$   | Remove duplicates from a sorted array in-place. Return the count of unique elements.                                                                              |
| 27  | ★ | Remove Element                         | $O(n) / O(1)$   | Remove all occurrences of a value from an array in-place. Return the new length.                                                                                  |
| 31  | ★ | Next Permutation                       |                 | Rearrange an array of integers to its lexicographically next greater permutation. If none, rearrange to lowest order.                                             |
| 41  | ★ | First Missing Positive                 |                 | Find the smallest missing positive integer in an unsorted array. Must run in $O(n)$ time and $O(1)$ space.                                                        |
| 42  | ★ | Trapping Rain Water                    | $O(n) / O(1)$   | Given an elevation map (array of bar heights), compute how much water it can trap after raining.                                                                  |
| 48  | ★ | Rotate Image                           |                 | Rotate an $n \times n$ matrix 90 degrees clockwise in-place.                                                                                                      |
| 54  | ★ | Spiral Matrix                          |                 | Return all elements of an $m \times n$ matrix in spiral order.                                                                                                    |
| 73  | ★ | Set Matrix Zeroes                      |                 | Given an $m \times n$ matrix, set entire row and column to 0 if any element is 0. Do it in-place.                                                                 |
| 75  | ★ | Sort Colors                            | $O(n) / O(1)$   | Sort an array of 0s, 1s, and 2s in-place without using sort (Dutch National Flag problem).                                                                        |
| 80  | ★ | Remove Duplicates from Sorted Array II | $O(n) / O(1)$   | Remove duplicates from a sorted array so each element appears at most twice in-place.                                                                             |
| 88  | ★ | Merge Sorted Array                     | $O(m+n) / O(1)$ | Merge two sorted integer arrays nums1 and nums2 into nums1 in-place (nums1 has extra space).                                                                      |
| 128 | ★ | Longest Consecutive Sequence           |                 | Find the length of the longest consecutive sequence of integers in an unsorted array. $O(n)$ required.                                                            |
| 164 | ★ | Maximum Gap                            | $O(n) / O(n)$   | Find the maximum gap between successive elements in a sorted form of an array. $O(n)$ required.                                                                   |
| 167 | ★ | Two Sum II - Input Array Is Sorted     |                 | Given a sorted array, find two numbers that add to a target (1-indexed). Exactly one solution.                                                                    |
| 169 | ★ | Majority Element                       |                 | Find the majority element that appears more than $n/2$ times.                                                                                                     |
| 189 | ★ | Rotate Array                           |                 | Rotate an array to the right by k positions in-place.                                                                                                             |
| 217 | ★ | Contains Duplicate                     |                 | Check if any value appears at least twice in an integer array.                                                                                                    |
| 219 | ★ | Contains Duplicate II                  | $O(n) / O(k)$   | Check if there are two distinct indices i and j such that $nums[i] == nums[j]$ and $i - j$ .                                                                      |
| 220 | ★ | Contains Duplicate III                 |                 | Check if there are two indices i and j such that $nums[i] - nums[j]$ and $i - j$ .                                                                                |
| 229 | ★ | Majority Element II                    |                 | Find all elements appearing more than $n/3$ times in an array.                                                                                                    |
| 238 | ★ | Product of Array Except Self           |                 | Return an array where $output[i]$ is the product of all elements except $nums[i]$ . No division.                                                                  |

| #   | ★ | Name                                     | Complexity           | Description                                                                                                    |
|-----|---|------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------|
| 259 | ★ | 3Sum Smaller                             |                      | Count triplets with sum smaller than a target (sorted array).                                                  |
| 280 | ★ | Wiggle Sort                              |                      | Rearrange an array in-place so it alternates peaks and valleys.                                                |
| 283 | ★ | Move Zeroes                              | $O(n) / O(1)$        | Move all zeroes in an array to the end while maintaining relative order of non-zero elements.                  |
| 289 | ★ | Game of Life                             |                      | Simulate Conway's Game of Life: update all cells simultaneously based on neighbor counts.                      |
| 303 | ★ | Range Sum Query - Immutable              |                      | Answer multiple range sum queries on a static array efficiently.                                               |
| 304 | ★ | Range Sum Query 2D - Immutable           |                      | Answer multiple 2D region sum queries on a static matrix efficiently.                                          |
| 307 | ★ | Range Sum Query - Mutable                |                      | Support range sum queries and point updates on an array.                                                       |
| 311 | ★ | Sparse Matrix Multiplication             |                      | Multiply two sparse matrices efficiently (skip zero elements).                                                 |
| 315 | ★ | Count of Smaller Numbers After Self      | $O(n \log n) / O(n)$ | For each element, count elements to its right that are smaller.                                                |
| 325 | ★ | Maximum Size Subarray Sum Equals k       | $O(n) / O(n)$        | Find the maximum length subarray with sum equal to k.                                                          |
| 327 | ★ | Count of Range Sum                       | $O(n \log n) / O(n)$ | Count range sums that lie in [lower, upper].                                                                   |
| 334 | ★ | Increasing Triplet Subsequence           |                      | Return true if there exists $i < j < k$ with $nums[i] < nums[j] < nums[k]$ .                                   |
| 349 | ★ | Intersection of Two Arrays               |                      | Find the intersection of two integer arrays (unique elements only).                                            |
| 350 | ★ | Intersection of Two Arrays II            |                      | Find the intersection of two integer arrays including duplicates.                                              |
| 370 | ★ | Range Addition                           |                      | Apply range updates <code>[start, end] += inc</code> then return the final array.                              |
| 384 | ★ | Shuffle an Array                         |                      | Shuffle an integer array so each permutation is equally likely.                                                |
| 414 | ★ | Third Maximum Number                     |                      | Find the third maximum distinct integer. Return the maximum if no third.                                       |
| 419 | ★ | Battleships in a Board                   |                      | Count battleships in a board ('X's forming horizontal or vertical lines).                                      |
| 442 | ★ | Find All Duplicates in an Array          |                      | Values in [1,n], each appears once or twice. Return the ones appearing twice. $O(n)$ time, $O(1)$ extra space. |
| 448 | ★ | Find All Numbers Disappeared in an Array |                      | Find all integers in range [1,n] that are missing from an array of size n.                                     |
| 454 | ★ | 4Sum II                                  |                      | Count quadruplets (i,j,k,l) such that $A[i]+B[j]+C[k]+D[l]==0$ .                                               |
| 457 | ★ | Circular Array Loop                      |                      | Detect a cycle of length > 1 with consistent direction in a circular array of jumps.                           |
| 485 | ★ | Max Consecutive Ones                     |                      | Find the maximum number of consecutive 1s in a binary array.                                                   |
| 493 | ★ | Reverse Pairs                            |                      | Count pairs (i,j) with $i < j$ and $nums[i] > 2 * nums[j]$ .                                                   |
| 498 | ★ | Diagonal Traverse                        |                      | Traverse an $m \times n$ matrix diagonally, alternating direction.                                             |
| 517 | ★ | Super Washing Machines                   |                      | Minimize the max number of moves to equalize dresses across machines (one dress to a neighbor per move).       |
| 523 | ★ | Continuous Subarray Sum                  | $O(n) / O(n)$        | Check if a subarray exists with sum divisible by k.                                                            |
| 525 | ★ | Contiguous Array                         |                      | Find the maximum length subarray with equal number of 0s and 1s.                                               |
| 532 | ★ | K-diff Pairs in an Array                 |                      | Count unique k-diff pairs (i,j) in an array where $nums[i]-nums[j]==k$ .                                       |
| 560 | ★ | Subarray Sum Equals K                    | $O(n) / O(n)$        | Count subarrays that sum to exactly k.                                                                         |
| 566 | ★ | Reshape the Matrix                       |                      | Reshape an $m \times n$ matrix to $r \times c$ without changing element order.                                 |
| 581 | ★ | Shortest Unsorted Continuous Subarray    |                      | Find the shortest subarray that, if sorted, makes the whole array sorted.                                      |

| #    | ★ | Name                                                        | Complexity             | Description                                                                                                 |
|------|---|-------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------|
| 594  | ★ | Longest Harmonious Subsequence                              |                        | Find the longest subsequence where max and min differ by exactly 1.                                         |
| 599  | ★ | Minimum Index Sum of Two Lists                              |                        | Find the common strings between two lists with the minimum index sum.                                       |
| 611  | ★ | Valid Triangle Number                                       |                        | Count the number of valid triangles from a sorted array of lengths.                                         |
| 661  | ★ | Image Smoother                                              |                        | Replace each cell with the floor of the average of itself and its up-to-8 neighbors.                        |
| 723  | ★ | Candy Crush                                                 |                        | Simulate Candy Crush: repeatedly remove groups of 3 same-color candies and drop.                            |
| 724  | ★ | Find Pivot Index                                            |                        | Find a pivot index where left sum equals right sum.                                                         |
| 766  | ★ | Toeplitz Matrix                                             |                        | Check if a matrix is Toeplitz (all diagonals have the same value).                                          |
| 769  | ★ | Max Chunks To Make Sorted                                   |                        | Array is a permutation of [0,n-1]. Max number of chunks that can be sorted independently to sort the whole. |
| 795  | ★ | Number of Subarrays with Bounded Maximum                    |                        | Count subarrays with a maximum value in range [L, R].                                                       |
| 807  | ★ | Max Increase to Keep City Skyline                           |                        | Maximize the total increase in building heights while preserving the city's skyline from all 4 directions.  |
| 825  | ★ | Friends Of Appropriate Ages                                 |                        | Count pairs of people where one person's age allows them to friend the other.                               |
| 838  | ★ | Push Dominoes                                               |                        | Given a string of dominoes ('L', 'R', '.'), simulate their fall.                                            |
| 845  | ★ | Longest Mountain in Array                                   |                        | Find the length of the longest mountain subarray (increases then decreases).                                |
| 896  | ★ | Monotonic Array                                             |                        | Check if an array is monotonically increasing or decreasing.                                                |
| 912  | ★ | Sort an Array                                               | $O(n \log n)$ / $O(n)$ | Sort an integer array. Implement an $O(n \log n)$ sort.                                                     |
| 974  | ★ | Subarray Sums Divisible by K                                | $O(n)$ / $O(k)$        | Count subarrays whose sum is divisible by k.                                                                |
| 977  | ★ | Squares of a Sorted Array                                   |                        | Return the squares of a sorted array in sorted order.                                                       |
| 1013 | ★ | Partition Array Into Three Parts With Equal Sum             |                        | Can the array be split into three contiguous parts with equal sums?                                         |
| 1074 | ★ | Number of Submatrices That Sum to Target                    |                        | Count submatrices summing to target.                                                                        |
| 1109 | ★ | Corporate Flight Bookings                                   |                        | Given bookings <code>[first, last, seats]</code> , return total seats booked per flight.                    |
| 1213 | ★ | Intersection of Three Sorted Arrays                         |                        | Find the intersection of three sorted arrays.                                                               |
| 1275 | ★ | Find Winner on a Tic Tac Toe Game                           |                        | Determine the winner of a Tic-Tac-Toe game given a sequence of moves.                                       |
| 1351 | ★ | Count Negative Numbers in a Sorted Matrix                   |                        | Count the number of negative numbers in a sorted matrix.                                                    |
| 1424 | ★ | Diagonal Traverse II                                        |                        | Traverse a matrix diagonally from bottom-left to top-right.                                                 |
| 1460 | ★ | Make Two Arrays Equal by Reversing Sub-arrays               |                        | Check if two arrays become equal after reversing any sub-array of one.                                      |
| 1480 | ★ | Running Sum of 1d Array                                     |                        | Return the running sum of an array.                                                                         |
| 1498 | ★ | Number of Subsequences That Satisfy the Given Sum Condition |                        | Count the number of subsequences where min+max target.                                                      |
| 1748 | ★ | Sum of Unique Elements                                      |                        | Find the sum of all unique elements in an array.                                                            |
| 1762 | ★ | Buildings With an Ocean View                                |                        | Find all buildings with an ocean view (nothing taller to their right).                                      |
| 1868 | ★ | Product of Two Run-Length Encoded Arrays                    |                        | Multiply two run-length encoded arrays.                                                                     |
| 1877 | ★ | Minimize Maximum Pair Sum in Array                          |                        | Minimize the maximum pair sum after optimally pairing elements.                                             |

| #    | ★ | Name                                             | Complexity | Description                                                                     |
|------|---|--------------------------------------------------|------------|---------------------------------------------------------------------------------|
| 1893 | ★ | Check if All the Integers in a Range Are Covered |            | Check if all integers in [left, right] are covered by at least one given range. |
| 1894 | ★ | Find the Student that Will Replace the Chalk     |            | Find the index of the student who will replace the chalk.                       |
| 1920 | ★ | Build Array from Permutation                     |            | Build an array where result[i] = nums[nums[i]].                                 |
| 1929 | ★ | Concatenation of Array                           |            | Return the concatenation of an array with itself.                               |

## Sliding Window (21 problems, 21 templated)

| #    | ★ | Name                                                   | Complexity    | Description                                                                                                         |
|------|---|--------------------------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------|
| 3    | ★ | Longest Substring Without Repeating Characters         | $O(n) / O(1)$ | Find the length of the longest substring without any repeating characters in a given string.                        |
| 30   | ★ | Substring with Concatenation of All Words              |               | Find all starting indices in s where a concatenation of all words in an array appears as a substring.               |
| 76   | ★ | Minimum Window Substring                               | $O(n) / O(k)$ | Find the minimum window substring of s that contains all characters in t.                                           |
| 159  | ★ | Longest Substring with At Most Two Distinct Characters | $O(n) / O(1)$ | Find the length of the longest substring containing at most two distinct characters.                                |
| 187  | ★ | Repeated DNA Sequences                                 |               | Find all 10-letter DNA sequences (substrings) that occur more than once.                                            |
| 209  | ★ | Minimum Size Subarray Sum                              |               | Find the minimal length subarray whose sum target. Return 0 if none.                                                |
| 239  | ★ | Sliding Window Maximum                                 | $O(n) / O(k)$ | Return the maximum value in each sliding window of size k across an array.                                          |
| 340  | ★ | Longest Substring with At Most K Distinct Characters   | $O(n) / O(k)$ | Find the length of the longest substring containing at most k distinct characters.                                  |
| 395  | ★ | Longest Substring with At Least K Repeating Characters |               | Find the longest substring where every character appears at least k times.                                          |
| 424  | ★ | Longest Repeating Character Replacement                | $O(n) / O(1)$ | Find the length of the longest substring where you can replace at most k characters to make all same.               |
| 438  | ★ | Find All Anagrams in a String                          | $O(n) / O(1)$ | Find all anagram start indices of pattern p in string s.                                                            |
| 487  | ★ | Max Consecutive Ones II                                |               | Given a binary array <code>nums</code> , return the maximum number of consecutive 1s if you may flip at most one 0. |
| 567  | ★ | Permutation in String                                  | $O(n) / O(1)$ | Check if any permutation of pattern p exists as a substring of string s.                                            |
| 643  | ★ | Maximum Average Subarray I                             |               | Find the maximum average of any subarray of length k.                                                               |
| 689  | ★ | Maximum Sum of 3 Non-Overlapping Subarrays             |               | Find 3 non-overlapping subarrays of length k with maximum total sum.                                                |
| 713  | ★ | Subarray Product Less Than K                           |               | Return the number of contiguous subarrays whose product of all elements is strictly less than <code>k</code> .      |
| 727  | ★ | Minimum Window Subsequence                             |               | Find the minimum window in s that contains all characters of t in order (subsequence).                              |
| 1004 | ★ | Max Consecutive Ones III                               | $O(n) / O(1)$ | Find the maximum consecutive 1s if you can flip at most k zeros.                                                    |
| 1044 | ★ | Longest Duplicate Substring                            |               | Find the longest duplicate substring using binary search and rolling hash.                                          |
| 1343 | ★ | #1343                                                  | $O(n) / O(1)$ | Count subarrays of size k with average $\geq$ threshold.                                                            |
| 1838 | ★ | Frequency of the Most Frequent Element                 |               | Find the minimum operations to make an array element appear the most frequently in a window.                        |

## Binary Search (30 problems, 30 templated)

| #    | ★ | Name                                                    | Complexity                    | Description                                                                                                       |
|------|---|---------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 4    | ★ | Median of Two Sorted Arrays                             |                               | Find the median of two sorted arrays of sizes m and n in $O(\log(m+n))$ time.                                     |
| 33   | ★ | Search in Rotated Sorted Array                          | $O(\log n) / O(1)$            | Search for a target in a sorted array that has been rotated at an unknown pivot. Return index or -1.              |
| 34   | ★ | Find First and Last Position of Element in Sorted Array | $O(\log n) / O(1)$            | Find the starting and ending positions of a target value in a sorted array. Return [-1, -1] if not found.         |
| 35   | ★ | Search Insert Position                                  | $O(\log n) / O(1)$            | Find the index where a target would be inserted in a sorted array to keep it sorted.                              |
| 69   | ★ | Sqrt(x)                                                 |                               | Compute the integer square root of a non-negative integer x without using sqrt().                                 |
| 74   | ★ | Search a 2D Matrix                                      |                               | Search for a target in an m×n matrix where each row is sorted and each row's first element > previous row's last. |
| 81   | ★ | Search in Rotated Sorted Array II                       |                               | Search in a sorted, rotated array that may contain duplicates.                                                    |
| 153  | ★ | Find Minimum in Rotated Sorted Array                    |                               | Find the minimum element in a sorted, rotated array with no duplicates.                                           |
| 162  | ★ | Find Peak Element                                       | $O(\log n) / O(1)$            | Find any peak element (element greater than its neighbors) in an array. $O(\log n)$ required.                     |
| 240  | ★ | Search a 2D Matrix II                                   |                               | Search for a target in an m×n matrix where rows and columns are both sorted.                                      |
| 278  | ★ | First Bad Version                                       |                               | Find the first bad version using a provided isBadVersion(n) API. Minimize API calls.                              |
| 374  | ★ | Guess Number Higher or Lower                            |                               | Guess the number between 1 and n using a provided guess() API. Minimize calls.                                    |
| 378  | ★ | Kth Smallest Element in a Sorted Matrix                 | $O(n \log n) / O(n)$          | Find the kth smallest element in an n×n matrix where rows and columns are sorted.                                 |
| 410  | ★ | Split Array Largest Sum                                 | $O(n \log \text{sum}) / O(1)$ | Split an array into m subarrays to minimize the largest subarray sum.                                             |
| 475  | ★ | Heaters                                                 |                               | Given house and heater positions, find the minimum heater radius so every house is covered by some heater.        |
| 540  | ★ | Single Element in a Sorted Array                        |                               | Find the single non-duplicate element in a sorted array where all others appear twice. $O(\log n)$ .              |
| 658  | ★ | Find K Closest Elements                                 |                               | Find k integers in a sorted array closest to x, ordered by closeness then value.                                  |
| 704  | ★ | Binary Search                                           | $O(\log n) / O(1)$            | Classic binary search: find target in sorted array.                                                               |
| 719  | ★ | Find K-th Smallest Pair Distance                        |                               | Find the k-th smallest absolute distance among all pairs in the array.                                            |
| 852  | ★ | Peak Index in a Mountain Array                          |                               | Find the peak index in a mountain array (element greater than both neighbors).                                    |
| 875  | ★ | Koko Eating Bananas                                     | $O(n \log \text{max}) / O(1)$ | Find the minimum eating speed k so Koko can eat all banana piles within h hours.                                  |
| 1011 | ★ | Capacity To Ship Packages Within D Days                 | $O(n \log \text{sum}) / O(1)$ | Find the minimum capacity to ship all packages within D days.                                                     |
| 1060 | ★ | Missing Element in Sorted Array                         |                               | Find the kth missing number in a sorted array.                                                                    |
| 1231 | ★ | #1231                                                   | $O(n \log \text{max}) / O(1)$ | Max min-piece when cutting chocolate into k pieces.                                                               |
| 1283 | ★ | #1283                                                   | $O(n \log \text{max}) / O(1)$ | Smallest divisor so sum of $\text{ceil}(\text{nums}[i]/\text{divisor}) \leq \text{threshold}$ .                   |

| #    | ★ | Name                                      | Complexity | Description                                                                   |
|------|---|-------------------------------------------|------------|-------------------------------------------------------------------------------|
| 1428 | ★ | Leftmost Column with at Least a One       |            | Find the leftmost column with at least one '1' in a binary matrix.            |
| 1482 | ★ | Minimum Number of Days to Make m Bouquets |            | Find the minimum number of days to make m bouquets of k adjacent flowers.     |
| 1539 | ★ | Kth Missing Positive Number               |            | Find the kth missing positive integer.                                        |
| 1818 | ★ | Minimum Absolute Sum Difference           |            | Find the minimum absolute sum difference after one substitution.              |
| 1891 | ★ | Cutting Ribbons                           |            | Find the maximum ribbon length such that m ribbons of that length can be cut. |

## Linked List (31 problems, 31 templated)

| #   | ★ | Name                                      | Complexity    | Description                                                                                                                                                        |
|-----|---|-------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   | ★ | Add Two Numbers                           | $O(n) / O(1)$ | Add two non-negative integers represented as reversed linked lists (ones digit first). Return the sum as a reversed linked list.                                   |
| 19  | ★ | Remove Nth Node From End of List          |               | Remove the nth node from the end of a linked list and return the head.                                                                                             |
| 21  | ★ | Merge Two Sorted Lists                    | $O(n) / O(1)$ | Merge two sorted linked lists into one sorted linked list in-place.                                                                                                |
| 24  | ★ | Swap Nodes in Pairs                       |               | Swap every two adjacent nodes in a linked list and return the modified head.                                                                                       |
| 25  | ★ | Reverse Nodes in k-Group                  | $O(n) / O(1)$ | Reverse nodes in a linked list k at a time. If remaining nodes < k, leave them as-is.                                                                              |
| 61  | ★ | Rotate List                               |               | Rotate a linked list to the right by k places.                                                                                                                     |
| 82  | ★ | Remove Duplicates from Sorted List II     | $O(n) / O(1)$ | Remove all nodes with duplicate values from a sorted linked list; keep only distinct values.                                                                       |
| 83  | ★ | Remove Duplicates from Sorted List        |               | Remove duplicate nodes from a sorted linked list; keep one copy of each value.                                                                                     |
| 86  | ★ | Partition List                            |               | Partition a linked list so all nodes with value < x come before nodes with value x.                                                                                |
| 92  | ★ | Reverse Linked List II                    |               | Reverse nodes m through n (1-indexed) of a linked list in one pass.                                                                                                |
| 109 | ★ | Convert Sorted List to Binary Search Tree |               | Convert a sorted linked list to a height-balanced BST.                                                                                                             |
| 138 | ★ | Copy List with Random Pointer             |               | Deep-copy a linked list where each node has a 'random' pointer to any node or null.                                                                                |
| 141 | ★ | Linked List Cycle                         | $O(n) / O(1)$ | Detect if a linked list has a cycle.                                                                                                                               |
| 142 | ★ | Linked List Cycle II                      | $O(n) / O(1)$ | Find the node where a cycle begins in a linked list.                                                                                                               |
| 143 | ★ | Reorder List                              |               | Reorder a linked list L01... to L01-1... in-place.                                                                                                                 |
| 148 | ★ | Sort List                                 |               | Sort a linked list in $O(n \log n)$ time and $O(1)$ extra space.                                                                                                   |
| 160 | ★ | Intersection of Two Linked Lists          | $O(n) / O(1)$ | Find the node where two singly linked lists intersect.                                                                                                             |
| 203 | ★ | Remove Linked List Elements               | $O(n) / O(1)$ | Remove all nodes with a given value from a linked list.                                                                                                            |
| 206 | ★ | Reverse Linked List                       |               | Reverse a singly linked list.                                                                                                                                      |
| 234 | ★ | Palindrome Linked List                    | $O(n) / O(1)$ | Check if a linked list is a palindrome in $O(n)$ time and $O(1)$ space.                                                                                            |
| 237 | ★ | Delete Node in a Linked List              |               | Delete a node from a singly linked list given only a reference to that node (not the head).                                                                        |
| 287 | ★ | Find the Duplicate Number                 | $O(n) / O(1)$ | Find the duplicate number in an array of n+1 integers in range [1,n]. $O(1)$ space required.                                                                       |
| 328 | ★ | Odd Even Linked List                      |               | Rearrange a linked list so all odd-indexed nodes come before even-indexed nodes.                                                                                   |
| 369 | ★ | Plus One Linked List                      |               | A non-negative integer is stored as a linked list of digits in big-endian order (most significant first). Add one to it and return the resulting list.             |
| 430 | ★ | Flatten a Multilevel Doubly Linked List   |               | Flatten a multilevel doubly linked list where nodes may have a child list.                                                                                         |
| 445 | ★ | Add Two Numbers II                        |               | Two numbers are stored as linked lists in big-endian order (most significant digit first). Add them and return the sum as a linked list, also in big-endian order. |
| 708 | ★ | Insert into a Sorted Circular Linked List |               | Insert a value into a sorted circular linked list.                                                                                                                 |
| 725 | ★ | Split Linked List in Parts                |               | Split a linked list into k consecutive parts as equal in size as possible.                                                                                         |
| 876 | ★ | Middle of the Linked List                 | $O(n) / O(1)$ | Find the middle node of a linked list (if two middles, return second).                                                                                             |

| #    | ★ | Name                                               | Complexity | Description                                                                   |
|------|---|----------------------------------------------------|------------|-------------------------------------------------------------------------------|
| 1171 | ★ | Remove Zero Sum Consecutive Nodes from Linked List |            | Remove consecutive nodes from a linked list whose values sum to zero.         |
| 1265 | ★ | Print Immutable Linked List in Reverse             |            | Print all values of an immutable linked list in reverse without modifying it. |

## String (87 problems, 87 templated)

| #   | ★ | Name                          | Complexity           | Description                                                                                                        |
|-----|---|-------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------|
| 6   | ★ | ZigZag Conversion             |                      | Convert a string into a zigzag pattern across numRows rows and read it row by row.                                 |
| 7   | ★ | Reverse Integer               | $O(\log n)$ / $O(1)$ | Reverse the digits of a 32-bit signed integer. Return 0 if the reversed value overflows.                           |
| 8   | ★ | String to Integer (atoi)      |                      | Implement atoi: parse a string, skip whitespace, handle optional sign, read digits, clamp to 32-bit integer range. |
| 12  | ★ | Integer to Roman              |                      | Convert an integer to its Roman numeral representation (values 1--3999).                                           |
| 13  | ★ | Roman to Integer              |                      | Convert a Roman numeral string to an integer.                                                                      |
| 14  | ★ | Longest Common Prefix         |                      | Find the longest common prefix string among an array of strings. Return '' if none.                                |
| 28  | ★ | Implement strStr()            |                      | Find the first occurrence of needle in haystack. Return -1 if not found.                                           |
| 38  | ★ | Count and Say                 |                      | Generate the nth term of the 'count and say' sequence: read digits of previous term aloud.                         |
| 43  | ★ | Multiply Strings              |                      | Multiply two non-negative integers given as strings. Return product as a string.                                   |
| 49  | ★ | Group Anagrams                | $O(nk)$ / $O(n)$     | Group an array of strings so that anagrams are together.                                                           |
| 58  | ★ | Length of Last Word           |                      | Return the length of the last word in a string (word = max sequence of non-space chars).                           |
| 65  | ★ | Valid Number                  |                      | Validate whether a string is a valid decimal number (integers, fractions, exponents).                              |
| 67  | ★ | Add Binary                    | $O(n)$ / $O(n)$      | Add two binary strings and return their binary sum as a string.                                                    |
| 68  | ★ | Text Justification            |                      | Format words into lines of maxWidth, fully justified (equal spaces, last line left-aligned).                       |
| 125 | ★ | Valid Palindrome              |                      | Check if a string is a palindrome, ignoring non-alphanumeric characters and case.                                  |
| 151 | ★ | Reverse Words in a String     |                      | Reverse the order of words in a string (split on spaces, handle multiple spaces).                                  |
| 161 | ★ | One Edit Distance             |                      | Determine if two strings are one edit distance apart (insert, delete, or replace one char).                        |
| 163 | ★ | Missing Ranges                |                      | Given a sorted array, return missing ranges between lower and upper bounds.                                        |
| 165 | ★ | Compare Version Numbers       |                      | Compare two version number strings (dot-separated integers). Return -1, 0, or 1.                                   |
| 166 | ★ | Fraction to Recurring Decimal |                      | Convert a fraction numerator/denominator to a decimal string, noting repeating parts in parentheses.               |
| 168 | ★ | Excel Sheet Column Title      | $O(\log n)$ / $O(1)$ | Convert an integer to its Excel column title (1, 26, 27, ...).                                                     |
| 171 | ★ | Excel Sheet Column Number     | $O(L)$ / $O(1)$      | Convert an Excel column title (A1, Z26, AA27) to its integer number.                                               |
| 179 | ★ | Largest Number                |                      | Arrange numbers so their concatenation forms the largest possible number.                                          |
| 186 | ★ | Reverse Words in a String II  |                      | Reverse words in a character array in-place (words separated by spaces).                                           |
| 205 | ★ | Isomorphic Strings            |                      | Check if two strings follow the same character-substitution pattern (isomorphic).                                  |
| 214 | ★ | Shortest Palindrome           |                      | Find the shortest palindrome by adding characters in front of a string.                                            |
| 228 | ★ | Summary Ranges                |                      | Given a sorted unique integer array, summarize consecutive runs as ranges.                                         |
| 242 | ★ | Valid Anagram                 | $O(n)$ / $O(1)$      | Determine if two strings are anagrams of each other.                                                               |
| 246 | ★ | Strobogrammatic Number        |                      | Check if a number reads the same when rotated 180 degrees.                                                         |
| 249 | ★ | Group Shifted Strings         |                      | Group strings that follow the same shift sequence (each char shifted by the same amount).                          |

| #   | ★ | Name                                        | Complexity  | Description                                                                                              |
|-----|---|---------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------|
| 266 | ★ | Palindrome Permutation                      |             | Check if a string can be rearranged into a palindrome (at most one odd-count char).                      |
| 271 | ★ | Encode and Decode Strings                   |             | Encode a list of strings to a single string and decode it back.                                          |
| 273 | ★ | Integer to English Words                    |             | Convert a non-negative integer to its English words representation.                                      |
| 290 | ★ | Word Pattern                                |             | Check if a string s follows a given pattern (bijective character mapping).                               |
| 299 | ★ | Bulls and Cows                              |             | Count bulls (right digit, right place) and cows (right digit, wrong place) in a number guessing game.    |
| 344 | ★ | Reverse String                              |             | Reverse a character array in-place.                                                                      |
| 345 | ★ | Reverse Vowels of a String                  |             | Reverse only the vowels of a string.                                                                     |
| 383 | ★ | Ransom Note                                 | O(n) / O(1) | Determine if a ransom note can be constructed from a magazine's letters.                                 |
| 387 | ★ | First Unique Character in a String          |             | Find the first non-repeating character in a string. Return its index or -1.                              |
| 388 | ★ | Longest Absolute File Path                  |             | Find the length of the longest absolute file path in a file system string.                               |
| 392 | ★ | Is Subsequence                              |             | Check if string s is a subsequence of string t.                                                          |
| 408 | ★ | Valid Word Abbreviation                     |             | Check if an abbreviation matches a word (digits represent skipped characters).                           |
| 409 | ★ | Longest Palindrome                          |             | Find the length of the longest palindrome that can be built from the given letters.                      |
| 415 | ★ | Add Strings                                 |             | Add two non-negative integers given as strings. Return the sum as a string.                              |
| 422 | ★ | Valid Word Square                           |             | Given a sequence of words, check whether they form a valid word square (kth row equals kth column).      |
| 423 | ★ | Reconstruct Original Digits from English    |             | Given a scrambled string of digit-word letters, decode original digits in order.                         |
| 434 | ★ | Number of Segments in a String              |             | Count the number of segments (maximal runs of non-space chars) in a string.                              |
| 443 | ★ | String Compression                          |             | Compress an array of chars in-place: 'aabcc' 'a2bc2'. Return new length.                                 |
| 459 | ★ | Repeated Substring Pattern                  |             | Check if a string can be built by repeating one of its substrings multiple times.                        |
| 468 | ★ | Validate IP Address                         |             | Validate whether a string is a valid IPv4 or IPv6 address.                                               |
| 500 | ★ | Keyboard Row                                |             | Find all words that can be typed on a single row of a QWERTY keyboard.                                   |
| 520 | ★ | Detect Capital                              |             | Check if a word is capitalized correctly (all caps, all lower, or first letter only).                    |
| 524 | ★ | Longest Word in Dictionary through Deleting |             | Find the longest word in a dictionary that is a subsequence of string s.                                 |
| 541 | ★ | Reverse String II                           |             | Reverse the first k characters of every 2k-character block in a string.                                  |
| 551 | ★ | Student Attendance Record I                 |             | Check if a student's attendance record is eligible for an award (no late > 1, no absence 3 consecutive). |
| 556 | ★ | Next Greater Element III                    |             | Find the next greater element by rearranging the digits of n. Return -1 if not possible or overflow.     |
| 557 | ★ | Reverse Words in a String III               |             | Reverse individual words in a string while preserving word order.                                        |
| 592 | ★ | Fraction Addition and Subtraction           |             | Add or subtract fractions, returning the result in lowest terms.                                         |
| 616 | ★ | Add Bold Tag in String                      |             | Wrap every substring of s that matches a word in the dictionary with <b>...</b>, merging overlaps.       |
| 657 | ★ | Robot Return to Origin                      |             | Check if a robot following instruction string (UDLR) returns to origin.                                  |
| 670 | ★ | Maximum Swap                                |             | Swap two digits in a number to get the maximum possible value.                                           |
| 680 | ★ | Valid Palindrome II                         |             | Check if a string is a palindrome after deleting at most one character.                                  |
| 681 | ★ | Next Closest Time                           |             | Find the next closest time using only the digits from a given time string.                               |
| 686 | ★ | Repeated String Match                       |             | Find the minimum number of times string a must repeat so b is a substring.                               |

| #    | ★ | Name                                   | Complexity | Description                                                                                                                              |
|------|---|----------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 709  | ★ | To Lower Case                          |            | Convert a string to lowercase.                                                                                                           |
| 726  | ★ | Number of Atoms                        |            | Parse a chemical formula string and return atom counts in sorted order.                                                                  |
| 748  | ★ | Shortest Completing Word               |            | Find the shortest word in a list that contains all letters of a license plate (case-insensitive, with multiplicity).                     |
| 788  | ★ | Rotated Digits                         |            | Count numbers in <code>[1, n]</code> that become a different valid number when each digit is rotated 180 degrees.                        |
| 791  | ★ | Custom Sort String                     |            | Sort string <code>s</code> so its characters appear in the order given by string <code>order</code> .                                    |
| 794  | ★ | Valid Tic-Tac-Toe State                |            | Decide whether the given tic-tac-toe board is reachable from valid play.                                                                 |
| 796  | ★ | Rotate String                          |            | Check if string <code>t</code> is a rotation of string <code>s</code> .                                                                  |
| 809  | ★ | Expressive Words                       |            | Determine if a query word matches a stretched version of a word in a list.                                                               |
| 824  | ★ | Goat Latin                             |            | Convert words to Goat Latin: append 'ma', move leading consonants to end + 'ma'.                                                         |
| 833  | ★ | Find And Replace in String             |            | Apply non-overlapping indexed replacements: at each index, if <code>sources[k]</code> matches, replace it with <code>targets[k]</code> . |
| 859  | ★ | Buddy Strings                          |            | Check if two strings are 'buddy strings': swapping exactly one pair makes them equal.                                                    |
| 953  | ★ | Verifying an Alien Dictionary          |            | Check if words are sorted in a given alien language's alphabetical order.                                                                |
| 1055 | ★ | Shortest Way to Form String            |            | Find the minimum number of subsequences of <code>source</code> whose concatenation equals <code>target</code> . Return -1 if impossible. |
| 1108 | ★ | Defanging an IP Address                |            | Defang an IP address by replacing '.' with '[.]'.                                                                                        |
| 1221 | ★ | Split a String in Balanced Strings     |            | Find the maximum number of balanced strings ('R' count == 'L' count) to split into.                                                      |
| 1328 | ★ | Break a Palindrome                     |            | Break a palindrome by changing one character to get the lexicographically smallest non-palindrome.                                       |
| 1392 | ★ | Longest Happy Prefix                   |            | Find the longest happy prefix (prefix that is also a suffix).                                                                            |
| 1446 | ★ | Consecutive Characters                 |            | Find the maximum number of consecutive same characters in a string.                                                                      |
| 1554 | ★ | Strings Differ by One Character        |            | Check if any two strings in a list differ by exactly one character.                                                                      |
| 1556 | ★ | Thousand Separator                     |            | Format an integer with a dot as the thousands separator.                                                                                 |
| 1736 | ★ | Latest Time by Replacing Hidden Digits |            | Find the latest valid time by replacing '?' with appropriate digits.                                                                     |
| 1816 | ★ | Truncate Sentence                      |            | Truncate a sentence to the first <code>k</code> words.                                                                                   |
| 1844 | ★ | Replace All Digits with Characters     |            | Replace each digit at odd positions with the corresponding letter.                                                                       |

## Stack / Queue / Heap (55 problems, 55 templated)

| #   | ★ | Name                             | Complexity           | Description                                                                                            |
|-----|---|----------------------------------|----------------------|--------------------------------------------------------------------------------------------------------|
| 20  | ★ | Valid Parentheses                | $O(n) / O(n)$        | Determine if a string of brackets '()' is valid --- every open bracket is closed in the correct order. |
| 23  | ★ | Merge k Sorted Lists             | $O(n \log k) / O(k)$ | Merge k sorted linked lists into one sorted linked list.                                               |
| 32  | ★ | Longest Valid Parentheses        |                      | Find the length of the longest valid (well-formed) parentheses substring.                              |
| 71  | ★ | Simplify Path                    |                      | Simplify an absolute Unix file path (handle '.', '..', multiple slashes).                              |
| 84  | ★ | Largest Rectangle in Histogram   | $O(n) / O(n)$        | Find the area of the largest rectangle that can be formed in a histogram.                              |
| 85  | ★ | Maximal Rectangle                | $O(mn) / O(n)$       | Find the largest rectangle containing only 1s in a binary matrix.                                      |
| 150 | ★ | Evaluate Reverse Polish Notation |                      | Evaluate a Reverse Polish Notation expression with +, -, *, /.                                         |
| 155 | ★ | Min Stack                        |                      | Design a stack that supports push, pop, top, and retrieving the minimum in $O(1)$ .                    |
| 215 | ★ | Kth Largest Element in an Array  | $O(n)$ avg / $O(1)$  | Find the kth largest element in an unsorted array (not necessarily distinct).                          |
| 218 | ★ | The Skyline Problem              |                      | Given building rectangles, return the skyline as a list of key points.                                 |
| 224 | ★ | Basic Calculator                 | $O(n) / O(n)$        | Evaluate a string expression with '+', '-', and nested parentheses.                                    |
| 225 | ★ | Implement Stack using Queues     |                      | Implement a LIFO stack using only queue operations.                                                    |
| 227 | ★ | Basic Calculator II              | $O(n) / O(n)$        | Evaluate a string expression with '+', '-', '*', '/' (no parentheses, integer division).               |
| 232 | ★ | Implement Queue using Stacks     |                      | Implement a queue using two stacks.                                                                    |
| 295 | ★ | Find Median from Data Stream     | $O(\log n) / O(n)$   | Design a data structure to find the median of a stream of integers efficiently.                        |
| 316 | ★ | Remove Duplicate Letters         |                      | Remove duplicate letters so each appears once, result is lexicographically smallest.                   |
| 347 | ★ | Top K Frequent Elements          | $O(n) / O(n)$        | Find the k most frequent elements in an integer array.                                                 |
| 373 | ★ | Find K Pairs with Smallest Sums  | $O(k \log k) / O(k)$ | Find the k pairs (u,v) with the smallest sums, one from each of two sorted arrays.                     |
| 394 | ★ | Decode String                    |                      | Decode a string encoded as k[encoded_string] where brackets can be nested.                             |
| 402 | ★ | Remove K Digits                  |                      | Remove k digits from a number string to get the smallest possible number.                              |
| 451 | ★ | Sort Characters By Frequency     |                      | Sort characters in a string by frequency of occurrence (descending).                                   |
| 456 | ★ | 132 Pattern                      |                      | Check if a 1-3-2 pattern exists in an integer array.                                                   |
| 480 | ★ | Sliding Window Median            | $O(n \log k) / O(k)$ | Find the median in each sliding window of size k across an array.                                      |
| 496 | ★ | Next Greater Element I           | $O(n) / O(n)$        | For each element in nums1, find the first greater element to its right in nums2.                       |
| 502 | ★ | IPO                              |                      | Maximize profit by choosing at most k projects, each requiring capital and providing profit.           |
| 503 | ★ | Next Greater Element II          | $O(n) / O(n)$        | Find the next greater element for each element in a circular array.                                    |
| 506 | ★ | Relative Ranks                   |                      | Assign ranks ("Gold Medal", "Silver Medal", "Bronze Medal", then 4, 5, ...) to athletes by score.      |
| 636 | ★ | Exclusive Time of Functions      |                      | Given a log of function start/end times, compute exclusive time per function.                          |

| #    | ★ | Name                                                | Complexity             | Description                                                                                |
|------|---|-----------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------|
| 678  | ★ | Valid Parenthesis String                            |                        | Check if a string with '(', ')', '*' (wildcard) can be a valid parenthesis string.         |
| 692  | ★ | Top K Frequent Words                                | $O(n \log k)$ / $O(n)$ | Find the k most frequent words (sorted by frequency, then lex).                            |
| 703  | ★ | Kth Largest Element in a Stream                     |                        | Design a class to find the kth largest element in a stream.                                |
| 716  | ★ | Max Stack                                           |                        | Design a stack with push, pop, top, getMax, and popMax operations.                         |
| 735  | ★ | Asteroid Collision                                  |                        | Simulate asteroids moving in a row: right-moving collide with left-moving.                 |
| 739  | ★ | Daily Temperatures                                  | $O(n)$ / $O(n)$        | Given temperatures, find the number of days until a warmer temperature.                    |
| 759  | ★ | Employee Free Time                                  |                        | Find the free time intervals common to all employees given their schedules.                |
| 767  | ★ | Reorganize String                                   | $O(n \log n)$ / $O(n)$ | Rearrange a string so no two adjacent characters are the same.                             |
| 772  | ★ | Basic Calculator III                                | $O(n)$ / $O(n)$        | Evaluate an arithmetic expression with +, -, *, /, and parentheses.                        |
| 844  | ★ | Backspace String Compare                            |                        | Check if two strings are equal after processing '#' as backspace.                          |
| 856  | ★ | Score of Parentheses                                |                        | Calculate the score of a valid parentheses string (empty=0, AB=A+B, (A)=2A or 1 if empty). |
| 862  | ★ | Shortest Subarray with Sum at Least K               |                        | Find the length of the shortest subarray with sum k (k can be negative).                   |
| 907  | ★ | Sum of Subarray Minimums                            |                        | Find the sum of subarray minimums for all subarrays.                                       |
| 921  | ★ | Minimum Add to Make Parentheses Valid               |                        | Find the minimum additions to make a parentheses string valid.                             |
| 946  | ★ | Validate Stack Sequences                            |                        | Check if a sequence can be produced by a series of push-pop operations on a stack.         |
| 973  | ★ | K Closest Points to Origin                          | $O(n)$ avg / $O(k)$    | Find the k points closest to the origin.                                                   |
| 1047 | ★ | Remove All Adjacent Duplicates In String            |                        | Remove all adjacent duplicate characters until no more adjacent duplicates exist.          |
| 1086 | ★ | High Five                                           |                        | For each student, compute the average of their top five scores.                            |
| 1190 | ★ | Reverse Substrings Between Each Pair of Parentheses |                        | Reverse the substrings between each pair of parentheses (innermost first).                 |
| 1209 | ★ | Remove All Adjacent Duplicates in String II         |                        | Remove all adjacent duplicates of length k in a string repeatedly.                         |
| 1249 | ★ | Minimum Remove to Make Valid Parentheses            |                        | Remove the minimum number of parentheses to make the string valid.                         |
| 1337 | ★ | The K Weakest Rows in a Matrix                      |                        | Find the k weakest rows in a matrix (fewest soldiers, ties broken by row index).           |
| 1541 | ★ | Minimum Insertions to Balance a Parentheses String  |                        | Find the minimum insertions to make a string balanced (every ')' needs two '(').           |
| 1614 | ★ | Maximum Nesting Depth of the Parentheses            |                        | Find the maximum nesting depth of parentheses in a string.                                 |
| 1792 | ★ | Maximum Average Pass Ratio                          |                        | Maximize the average pass ratio by adding extra students optimally.                        |
| 1944 | ★ | Number of Visible People in a Queue                 |                        | Count the number of people each person can see in a queue (taller blocks view).            |
| 1985 | ★ | Find the Kth Largest Integer in the Array           |                        | Find the kth largest integer in an array of number strings.                                |

## DFS / Backtracking (140 problems, 140 templated)

| #   | ★ | Name                                                       | Complexity        | Description                                                                                                       |
|-----|---|------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------|
| 17  | ★ | Letter Combinations of a Phone Number                      |                   | Given a string of digits 2-9, return all possible letter combinations a phone keypad could represent.             |
| 22  | ★ | Generate Parentheses                                       |                   | Generate all combinations of n pairs of well-formed parentheses.                                                  |
| 36  | ★ | Valid Sudoku                                               |                   | Determine if a 9×9 Sudoku board is valid (rows, columns, and 3×3 boxes each contain 1-9 with no repeats).         |
| 37  | ★ | Sudoku Solver                                              | $O(9^n) / O(1)$   | Solve a Sudoku puzzle by filling empty cells with digits 1-9 such that each row, column, and 3×3 box is valid.    |
| 39  | ★ | Combination Sum                                            | $O(2^n) / O(n)$   | Find all unique combinations of candidates (may reuse) that sum to a target.                                      |
| 40  | ★ | Combination Sum II                                         | $O(2^n) / O(n)$   | Find all unique combinations of candidates (each used once) that sum to a target. Candidates may have duplicates. |
| 46  | ★ | Permutations                                               | $O(n!) / O(n)$    | Return all possible permutations of a distinct-integer array.                                                     |
| 47  | ★ | Permutations II                                            | $O(n!) / O(n)$    | Return all unique permutations of an array that may contain duplicates.                                           |
| 51  | ★ | N-Queens                                                   | $O(n!) / O(n)$    | Place n queens on an n×n chessboard so no two attack each other. Return all distinct solutions.                   |
| 60  | ★ | Permutation Sequence                                       | $O(n) / O(n)$     | Find the kth permutation sequence of numbers 1 to n.                                                              |
| 77  | ★ | Combinations                                               |                   | Return all combinations of k numbers from 1 to n.                                                                 |
| 78  | ★ | Subsets                                                    | $O(2^n) / O(n)$   | Return all possible subsets (power set) of a distinct-integer array.                                              |
| 79  | ★ | Word Search                                                | $O(mn4^L) / O(L)$ | Search for a word in an m×n grid of characters by adjacent (up/down/left/right) traversal without reuse.          |
| 90  | ★ | Subsets II                                                 | $O(2^n) / O(n)$   | Return all unique subsets of an array that may contain duplicates.                                                |
| 93  | ★ | Restore IP Addresses                                       |                   | Restore all possible valid IP address strings from a string of digits.                                            |
| 94  | ★ | Binary Tree Inorder Traversal                              |                   | Return the inorder traversal (left--root--right) of a binary tree.                                                |
| 95  | ★ | Unique Binary Search Trees II                              |                   | Generate all structurally unique BSTs with values 1..n.                                                           |
| 98  | ★ | Validate Binary Search Tree                                | $O(n) / O(h)$     | Validate whether a binary tree is a valid Binary Search Tree.                                                     |
| 99  | ★ | Recover Binary Search Tree                                 |                   | Recover a BST where exactly two nodes have been swapped.                                                          |
| 100 | ★ | Same Tree                                                  |                   | Check if two binary trees are structurally identical with the same node values.                                   |
| 101 | ★ | Symmetric Tree                                             |                   | Check if a binary tree is symmetric (mirror image of itself).                                                     |
| 102 | ★ | Binary Tree Level Order Traversal                          | $O(n) / O(n)$     | Return node values level by level (left to right) in a binary tree.                                               |
| 103 | ★ | Binary Tree Zigzag Level Order Traversal                   | $O(n) / O(n)$     | Return node values in zigzag level order (left-to-right for odd levels, right-to-left for even).                  |
| 104 | ★ | Maximum Depth of Binary Tree                               | $O(n) / O(h)$     | Find the maximum depth (number of nodes along the longest root-to-leaf path) of a binary tree.                    |
| 105 | ★ | Construct Binary Tree from Preorder and Inorder Traversal  |                   | Reconstruct a binary tree from its preorder and inorder traversal arrays.                                         |
| 106 | ★ | Construct Binary Tree from Inorder and Postorder Traversal |                   | Reconstruct a binary tree from its inorder and postorder traversal arrays.                                        |
| 107 | ★ | Binary Tree Level Order Traversal II                       |                   | Return node values level by level from bottom to top.                                                             |
| 108 | ★ | Convert Sorted Array to Binary Search Tree                 |                   | Convert a sorted integer array to a height-balanced BST.                                                          |
| 110 | ★ | Balanced Binary Tree                                       | $O(n) / O(h)$     | Determine if a binary tree is height-balanced (every node's subtrees differ in depth by at most 1).               |
| 111 | ★ | Minimum Depth of Binary Tree                               | $O(n) / O(h)$     | Find the minimum depth of a binary tree (shortest path from root to a leaf).                                      |

| #   | ★ | Name                                           | Complexity       | Description                                                                                              |
|-----|---|------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------|
| 112 | ★ | Path Sum                                       |                  | Check if a root-to-leaf path exists whose node values sum to a given target.                             |
| 113 | ★ | Path Sum II                                    |                  | Find all root-to-leaf paths whose values sum to a given target.                                          |
| 114 | ★ | Flatten Binary Tree to Linked List             |                  | Flatten a binary tree into a linked list in-place (preorder, using right pointers).                      |
| 116 | ★ | Populating Next Right Pointers in Each Node    |                  | Populate each node's 'next' pointer to its right neighbor (perfect binary tree). Use O(1) extra space.   |
| 117 | ★ | Populating Next Right Pointers in Each Node II |                  | Same as 116 but for an arbitrary binary tree.                                                            |
| 124 | ★ | Binary Tree Maximum Path Sum                   | O(n) / O(h)      | Find the maximum path sum in a binary tree (path can start/end at any node).                             |
| 129 | ★ | Sum Root to Leaf Numbers                       |                  | Sum all root-to-leaf numbers where each path forms a number (root digit is most significant).            |
| 144 | ★ | Binary Tree Preorder Traversal                 |                  | Return the preorder traversal (root → left → right) of a binary tree.                                    |
| 145 | ★ | Binary Tree Postorder Traversal                |                  | Return the postorder traversal (left → right → root) of a binary tree.                                   |
| 199 | ★ | Binary Tree Right Side View                    | O(n) / O(n)      | Return the values visible from the right side of a binary tree (one per level).                          |
| 216 | ★ | Combination Sum III                            | O(C(9,k)) / O(k) | Find all combinations of k distinct numbers from 1--9 that sum to n.                                     |
| 222 | ★ | Count Complete Tree Nodes                      |                  | Count nodes in a complete binary tree without visiting all nodes. Faster than O(n).                      |
| 226 | ★ | Invert Binary Tree                             |                  | Invert (mirror) a binary tree.                                                                           |
| 230 | ★ | Kth Smallest Element in a BST                  | O(n) / O(h)      | Find the kth smallest element in a BST.                                                                  |
| 235 | ★ | Lowest Common Ancestor of a Binary Search Tree |                  | Find the lowest common ancestor of two nodes in a BST.                                                   |
| 236 | ★ | Lowest Common Ancestor of a Binary Tree        |                  | Find the lowest common ancestor of two nodes in a binary tree (not necessarily a BST).                   |
| 241 | ★ | Different Ways to Add Parentheses              |                  | Given an expression string, compute all possible results from different ways to parenthesize it.         |
| 250 | ★ | Count Unival Subtrees                          |                  | Count subtrees in which all nodes share the same value.                                                  |
| 255 | ★ | Verify Preorder Sequence in Binary Search Tree |                  | Verify whether an array of integers is the correct preorder traversal of a BST.                          |
| 257 | ★ | Binary Tree Paths                              |                  | Return all root-to-leaf paths in a binary tree as strings ('1->2->5').                                   |
| 267 | ★ | Palindrome Permutation II                      |                  | Return all palindrome permutations of a string.                                                          |
| 270 | ★ | Closest Binary Search Tree Value               | O(h) / O(1)      | Find the value in a BST closest to a given target.                                                       |
| 272 | ★ | Closest Binary Search Tree Value II            |                  | Find k values in a BST closest to target.                                                                |
| 282 | ★ | Expression Add Operators                       |                  | Insert operators +, -, * between digits of a string num to reach a target value. Return all expressions. |
| 285 | ★ | Inorder Successor in BST                       | O(h) / O(1)      | Find the inorder successor of a given node in a BST.                                                     |
| 291 | ★ | Word Pattern II                                |                  | Check if a string s follows a given pattern where pattern chars map to non-overlapping substrings.       |
| 293 | ★ | Flip Game                                      |                  | Given a string of '+' and '-', return all states after flipping one "++" to "--".                        |
| 294 | ★ | Flip Game II                                   |                  | Determine if the first player can guarantee a win in Flip Game.                                          |
| 297 | ★ | Serialize and Deserialize Binary Tree          |                  | Serialize and deserialize a binary tree to/from a string.                                                |
| 298 | ★ | Binary Tree Longest Consecutive Sequence       |                  | Find the longest consecutive sequence in a binary tree (parent to child, values increasing by 1).        |
| 301 | ★ | Remove Invalid Parentheses                     |                  | Remove the minimum number of parentheses to make the string valid. Return all results.                   |

| #   | ★ | Name                                                    | Complexity    | Description                                                                                        |
|-----|---|---------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------|
| 306 | ★ | Additive Number                                         |               | Check if a string forms an additive number (each number = sum of two preceding).                   |
| 314 | ★ | Binary Tree Vertical Order Traversal                    |               | Return the binary tree's node values ordered by column, then by row.                               |
| 320 | ★ | Generalized Abbreviation                                |               | Return all possible abbreviations of a word.                                                       |
| 339 | ★ | Nested List Weight Sum                                  |               | Compute the weighted sum of a nested integer list where weight equals depth.                       |
| 364 | ★ | Nested List Weight Sum II                               |               | Sum each integer weighted by its inverse depth (deepest integers have weight 1).                   |
| 386 | ★ | Lexicographical Numbers                                 |               | Return integers 1..n in lexicographical order.                                                     |
| 404 | ★ | Sum of Left Leaves                                      |               | Find the sum of all left leaves in a binary tree.                                                  |
| 426 | ★ | Convert Binary Search Tree to Sorted Doubly Linked List |               | Convert a BST to a sorted circular doubly linked list in-place.                                    |
| 429 | ★ | N-ary Tree Level Order Traversal                        |               | Return the level order traversal of an N-ary tree.                                                 |
| 437 | ★ | Path Sum III                                            | $O(n) / O(n)$ | Count paths in a binary tree that sum to a given value (path goes downward, any node to any node). |
| 440 | ★ | K-th Smallest in Lexicographical Order                  |               | Find the kth smallest integer in 1..n by lexicographical order.                                    |
| 449 | ★ | Serialize and Deserialize BST                           |               | Serialize and deserialize a BST compactly.                                                         |
| 450 | ★ | Delete Node in a BST                                    |               | Delete a given key from a BST and return the updated root.                                         |
| 473 | ★ | Matchsticks to Square                                   |               | Determine if n matchsticks can form exactly 4 equal-length sides of a square.                      |
| 488 | ★ | Zuma Game                                               |               | Find the minimum number of marbles to insert in Zuma to clear the board.                           |
| 489 | ★ | Robot Room Cleaner                                      |               | Design a robot cleaner: given room-cleaning robot with no map, clean the entire room.              |
| 491 | ★ | Increasing Subsequences                                 |               | Return all increasing subsequences of length $\geq 2$ (input may have duplicates).                 |
| 501 | ★ | Find Mode in Binary Search Tree                         |               | Find the mode(s) in a BST (values that appear most frequently).                                    |
| 508 | ★ | Most Frequent Subtree Sum                               |               | Find the subtree sum(s) that occur most frequently.                                                |
| 510 | ★ | Inorder Successor in BST II                             |               | Find the inorder successor of a node given only a parent pointer (no root).                        |
| 513 | ★ | #513                                                    | $O(n) / O(n)$ | Find leftmost value in last row of binary tree.                                                    |
| 515 | ★ | Find Largest Value in Each Tree Row                     |               | Return the maximum value found at each level.                                                      |
| 526 | ★ | Beautiful Arrangement                                   |               | Count 'beautiful arrangements': n numbers placed so each divides or is divisible by its position.  |
| 536 | ★ | Construct Binary Tree from String                       |               | Construct a binary tree from its string representation '4(2(3)(1))(6(5))'.                         |
| 538 | ★ | Convert BST to Greater Tree                             |               | Convert each BST node's value to the sum of all values greater than or equal to it.                |
| 543 | ★ | Diameter of Binary Tree                                 | $O(n) / O(h)$ | Find the diameter of a binary tree (longest path between any two nodes).                           |
| 545 | ★ | Boundary of Binary Tree                                 |               | Return the boundary of a binary tree: left boundary + leaves + right boundary (no duplicates).     |
| 549 | ★ | Binary Tree Longest Consecutive Sequence II             | $O(n) / O(h)$ | Find the longest consecutive sequence in a binary tree (can go up or down, must be consecutive).   |
| 559 | ★ | Maximum Depth of N-ary Tree                             |               | Find the maximum depth of an N-ary tree.                                                           |
| 563 | ★ | Binary Tree Tilt                                        |               | Find the total tilt of a binary tree (sum of leftSum - rightSum for all nodes).                    |
| 572 | ★ | Subtree of Another Tree                                 |               | Check if tree t is a subtree of tree s.                                                            |
| 589 | ★ | N-ary Tree Preorder Traversal                           |               | Return the preorder traversal of an N-ary tree.                                                    |
| 590 | ★ | N-ary Tree Postorder Traversal                          |               | Return the postorder traversal of an N-ary tree.                                                   |

| #   | ★ | Name                                                        | Complexity      | Description                                                                                                                                                                        |
|-----|---|-------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 606 | ★ | Construct String from Binary Tree                           |                 | Construct a string from a binary tree using preorder with parentheses.                                                                                                             |
| 617 | ★ | Merge Two Binary Trees                                      |                 | Merge two binary trees by summing overlapping nodes.                                                                                                                               |
| 637 | ★ | Average of Levels in Binary Tree                            |                 | Return the average value of each level of a binary tree.                                                                                                                           |
| 652 | ★ | Find Duplicate Subtrees                                     |                 | Find all duplicate subtrees in a binary tree; return root of each.                                                                                                                 |
| 653 | ★ | Two Sum IV - Input is a BST                                 |                 | Find two numbers in a BST that sum to a given target.                                                                                                                              |
| 654 | ★ | Maximum Binary Tree                                         |                 | Build the maximum binary tree: root is array max, left from left subarray, right from right.                                                                                       |
| 662 | ★ | Maximum Width of Binary Tree                                | $O(n)$ / $O(n)$ | Return the maximum width of any level of a binary tree. Width is defined as the length between the leftmost and rightmost non-null nodes, including all the null nodes in between. |
| 671 | ★ | Second Minimum Node In a Binary Tree                        |                 | In a tree where each node's value is the min of its children, find the second smallest value overall.                                                                              |
| 679 | ★ | 24 Game                                                     |                 | Check if 4 numbers with any order and any of +, -, *, / can produce 24.                                                                                                            |
| 687 | ★ | #687                                                        | $O(n)$ / $O(h)$ | Longest univalue path --- edges between nodes of equal value.                                                                                                                      |
| 698 | ★ | Partition to K Equal Sum Subsets                            |                 | Determine if an array can be split into k subsets of equal sum.                                                                                                                    |
| 700 | ★ | Search in a Binary Search Tree                              |                 | Search for a value in a BST.                                                                                                                                                       |
| 701 | ★ | Insert into a Binary Search Tree                            |                 | Insert a value into a BST and return the updated root.                                                                                                                             |
| 742 | ★ | Closest Leaf in a Binary Tree                               |                 | Find the node in a binary tree closest to the target that is a leaf.                                                                                                               |
| 776 | ★ | Split BST                                                   |                 | Split a BST into two trees: one with values $\leq V$ and one with values $> V$ .                                                                                                   |
| 784 | ★ | Letter Case Permutation                                     |                 | Return all strings formed by toggling the case of letters in a string.                                                                                                             |
| 814 | ★ | Binary Tree Pruning                                         |                 | Prune a binary tree by removing subtrees containing no 1s.                                                                                                                         |
| 842 | ★ | Split Array into Fibonacci Sequence                         |                 | Check if a string can be split into a Fibonacci-like sequence.                                                                                                                     |
| 863 | ★ | All Nodes Distance K in Binary Tree                         |                 | Find all nodes at distance k from a target node in a binary tree.                                                                                                                  |
| 865 | ★ | Smallest Subtree with all the Deepest Nodes                 |                 | Find the smallest subtree containing all the deepest leaves.                                                                                                                       |
| 889 | ★ | Construct Binary Tree from Preorder and Postorder Traversal |                 | Reconstruct a binary tree from preorder and postorder traversals (any valid tree).                                                                                                 |
| 897 | ★ | Increasing Order Search Tree                                |                 | Transform a BST into an increasing-order search tree (right-skewed).                                                                                                               |
| 919 | ★ | Complete Binary Tree Inserter                               |                 | Design a complete binary tree inserter with $O(1)$ insert.                                                                                                                         |
| 932 | ★ | Beautiful Array                                             |                 | Construct a beautiful array where no three elements form an arithmetic progression.                                                                                                |
| 938 | ★ | Range Sum of BST                                            |                 | Return the sum of all BST node values within the range [low, high].                                                                                                                |
| 951 | ★ | Flip Equivalent Binary Trees                                |                 | Check if two binary trees are flip-equivalent (can make them identical by flipping children).                                                                                      |
| 958 | ★ | Check Completeness of a Binary Tree                         |                 | Check if a binary tree is a complete binary tree.                                                                                                                                  |
| 987 | ★ | Vertical Order Traversal of a Binary Tree                   |                 | Return node values grouped by vertical position (column), then by row, then by value.                                                                                              |
| 988 | ★ | Smallest String Starting From Leaf                          |                 | Find the lexicographically smallest string from root-to-leaf path.                                                                                                                 |
| 993 | ★ | Cousins in Binary Tree                                      |                 | Check if two nodes in a binary tree are cousins (same depth, different parents).                                                                                                   |
| 998 | ★ | Maximum Binary Tree II                                      |                 | Insert a value into a maximum tree built by appending the value to the original array's end.                                                                                       |

| #    | ★ | Name                                                 | Complexity | Description                                                                         |
|------|---|------------------------------------------------------|------------|-------------------------------------------------------------------------------------|
| 1008 | ★ | Construct Binary Search Tree from Preorder Traversal |            | Construct a BST from its preorder traversal.                                        |
| 1026 | ★ | Maximum Difference Between Node and Ancestor         |            | Find the maximum difference between a node and its ancestor in a binary tree.       |
| 1038 | ★ | Binary Search Tree to Greater Sum Tree               |            | Convert each BST node to the sum of all nodes with a greater or equal value.        |
| 1087 | ★ | Brace Expansion                                      |            | Return all strings formed by expanding brace expressions (comma-separated choices). |
| 1104 | ★ | Path In Zigzag Labelled Binary Tree                  |            | Find the parent node in a zigzag-labeled infinite binary tree.                      |
| 1110 | ★ | Delete Nodes And Return Forest                       |            | Delete given nodes from a binary tree and return the roots of the remaining forest. |
| 1123 | ★ | Lowest Common Ancestor of Deepest Leaves             |            | Find the LCA of the deepest leaves in a binary tree.                                |
| 1161 | ★ | Maximum Level Sum of a Binary Tree                   |            | Find the level of a binary tree with the maximum sum of node values.                |
| 1214 | ★ | Two Sum BSTs                                         |            | Given two BSTs and a target, decide if a value from each sums to the target.        |
| 1305 | ★ | All Elements in Two Binary Search Trees              |            | Return all elements from two BSTs in sorted order.                                  |
| 1361 | ★ | Validate Binary Tree Nodes                           |            | Given child arrays, verify the nodes form exactly one valid binary tree.            |
| 1382 | ★ | Balance a Binary Search Tree                         |            | Convert a BST to a balanced BST.                                                    |
| 1522 | ★ | Diameter of N-Ary Tree                               |            | Find the diameter of an N-ary tree.                                                 |
| 1644 | ★ | Lowest Common Ancestor of a Binary Tree II           |            | Find the LCA of two nodes in a binary tree where nodes may not exist.               |
| 1650 | ★ | Lowest Common Ancestor of a Binary Tree III          |            | Find the LCA of two nodes given parent pointers (no root access).                   |

## Graph (58 problems, 58 templated)

| #   | ★ | Name                                                  | Complexity        | Description                                                                                                                                            |
|-----|---|-------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 126 | ★ | Word Ladder II                                        |                   | Find all shortest transformation sequences from beginWord to endWord, changing one letter at a time.                                                   |
| 127 | ★ | Word Ladder                                           | $O(V+E) / O(V)$   | Find the length of the shortest transformation sequence from beginWord to endWord (one letter at a time, each word must be in wordList).               |
| 130 | ★ | Surrounded Regions                                    | $O(mn) / O(mn)$   | Capture all 'O' regions not connected to the border (surrounded by 'X'); flip them to 'X'.                                                             |
| 133 | ★ | Clone Graph                                           |                   | Deep-copy (clone) an undirected graph where each node has a value and a neighbors list.                                                                |
| 200 | ★ | Number of Islands                                     | $O(mn) / O(mn)$   | Count the number of islands (connected regions of '1's) in a 2D binary grid.                                                                           |
| 207 | ★ | Course Schedule                                       | $O(V+E) / O(V+E)$ | Determine if all n courses can be finished given prerequisite pairs (detect cycle in directed graph).                                                  |
| 210 | ★ | Course Schedule II                                    | $O(V+E) / O(V+E)$ | Return a valid course order given prerequisites (topological sort). Return [] if impossible.                                                           |
| 261 | ★ | Graph Valid Tree                                      | $O(V+E) / O(V)$   | Determine if n nodes and given edges form a valid undirected tree.                                                                                     |
| 269 | ★ | Alien Dictionary                                      | $O(V+E) / O(V+E)$ | Given a sorted list of alien words, determine the order of characters in the alien alphabet. Return the order as a string, or empty string if invalid. |
| 277 | ★ | Find the Celebrity                                    |                   | Find the celebrity in a group of n people: everyone knows them, they know nobody.                                                                      |
| 286 | ★ | Walls and Gates                                       | $O(mn) / O(mn)$   | Walls and Gates: fill each empty room (INF) with its distance to the nearest gate (0).                                                                 |
| 296 | ★ | Best Meeting Point                                    |                   | Find the point minimizing total Manhattan distance to a set of buildings on a grid.                                                                    |
| 305 | ★ | Number of Islands II                                  |                   | Count islands after each cell is added (online algorithm).                                                                                             |
| 310 | ★ | Minimum Height Trees                                  | $O(V+E) / O(V+E)$ | Find all minimum height trees in an undirected tree graph.                                                                                             |
| 317 | ★ | Shortest Distance from All Buildings                  |                   | Find the empty land cell minimizing total walking distance to all buildings.                                                                           |
| 323 | ★ | Number of Connected Components in an Undirected Graph | $O(V+E) / O(V)$   | Count connected components in an undirected graph.                                                                                                     |
| 329 | ★ | Longest Increasing Path in a Matrix                   |                   | Find the length of the longest increasing path in a matrix (4-directional movement).                                                                   |
| 399 | ★ | Evaluate Division                                     |                   | Given equations and ratios, evaluate new queries (transitive division).                                                                                |
| 407 | ★ | Trapping Rain Water II                                |                   | Given a 2D height map, compute how much water it can trap after raining.                                                                               |
| 417 | ★ | Pacific Atlantic Water Flow                           | $O(mn) / O(mn)$   | Find all cells from which water can flow to both Pacific and Atlantic oceans (4-directional).                                                          |
| 463 | ★ | Island Perimeter                                      |                   | Calculate the perimeter of islands in a grid.                                                                                                          |
| 529 | ★ | Minesweeper                                           |                   | Simulate the first click in Minesweeper: reveal cells per the game rules.                                                                              |
| 542 | ★ | 01 Matrix                                             |                   | For each cell in a binary matrix, find the distance to the nearest 0.                                                                                  |
| 547 | ★ | Number of Provinces                                   |                   | Count the number of provinces (connected components) among n cities given friendship pairs.                                                            |
| 684 | ★ | #684                                                  | $O(E) / O(V)$     | Remove one redundant edge to make graph a tree.                                                                                                        |
| 694 | ★ | Number of Distinct Islands                            |                   | Count the number of distinct islands in a binary grid (by shape, not position).                                                                        |
| 695 | ★ | Max Area of Island                                    | $O(mn) / O(mn)$   | Find the maximum area of any island in a binary grid.                                                                                                  |
| 721 | ★ | #721                                                  | $O(n^2) / O(n)$   | Merge email accounts sharing a common email address.                                                                                                   |

| #    | ★ | Name                                               | Complexity                 | Description                                                                                                                                                              |
|------|---|----------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 733  | ★ | Flood Fill                                         |                            | Flood-fill a pixel in an image: change all connected same-color pixels.                                                                                                  |
| 743  | ★ | Network Delay Time                                 | $O((V+E) \log V) / O(V+E)$ | Find the minimum time for a signal to reach all nodes in a network (Dijkstra).                                                                                           |
| 752  | ★ | Open the Lock                                      |                            | Find the minimum number of turns to open a 4-digit lock from '0000', avoiding dead-ends.                                                                                 |
| 778  | ★ | Swim in Rising Water                               |                            | Find the minimum time to swim from top-left to bottom-right of a grid (elevation = wait time).                                                                           |
| 785  | ★ | Is Graph Bipartite?                                |                            | Check if an undirected graph can be 2-colored (bipartite).                                                                                                               |
| 797  | ★ | All Paths From Source to Target                    |                            | Find all paths from node 0 to node n-1 in a DAG.                                                                                                                         |
| 802  | ★ | Find Eventual Safe States                          |                            | Find all safe nodes (nodes from which all paths lead to terminal nodes) in a DAG.                                                                                        |
| 815  | ★ | Bus Routes                                         |                            | Find the minimum number of buses to take to travel from source to target stop.                                                                                           |
| 827  | ★ | Making A Large Island                              | $O(mn) / O(mn)$            | Connect two islands to maximize island size; return the maximum.                                                                                                         |
| 847  | ★ | Shortest Path Visiting All Nodes                   |                            | Find the shortest path visiting all nodes in an undirected graph.                                                                                                        |
| 851  | ★ | Loud and Rich                                      |                            | Find the quietest person among those richer than each person.                                                                                                            |
| 886  | ★ | Possible Bipartition                               |                            | Check if people can be split into two groups with no one in the same group disliking each other.                                                                         |
| 909  | ★ | Snakes and Ladders                                 | $O(n^2) / O(n^2)$          | Find the minimum number of dice rolls to reach the last square (snakes and ladders).                                                                                     |
| 913  | ★ | Cat and Mouse                                      |                            | A mouse (start node 1) and cat (start node 2) move on a graph; mouse wins by reaching node 0, cat wins by catching the mouse, else draw. Return 0/1/2 with optimal play. |
| 934  | ★ | Shortest Bridge                                    |                            | Find the shortest bridge between two islands in a binary matrix.                                                                                                         |
| 994  | ★ | Rotting Oranges                                    | $O(mn) / O(mn)$            | Find the minimum time to rot all fresh oranges (multi-source BFS).                                                                                                       |
| 1034 | ★ | Coloring A Border                                  |                            | Color cells on the border of an island (connected component) with a given color.                                                                                         |
| 1091 | ★ | Shortest Path in Binary Matrix                     | $O(mn) / O(mn)$            | Find the shortest path from top-left to bottom-right in a binary matrix (8-directional).                                                                                 |
| 1102 | ★ | Path With Maximum Minimum Value                    |                            | Find the path from top-left to bottom-right maximizing the minimum value on the path.                                                                                    |
| 1136 | ★ | Parallel Courses                                   |                            | Find the minimum number of months to complete all courses given prerequisites and durations.                                                                             |
| 1162 | ★ | As Far from Land as Possible                       |                            | In a grid of <b>0</b> (water) and <b>1</b> (land), find the water cell whose distance to the nearest land is maximized; return that distance, or -1.                     |
| 1197 | ★ | Minimum Knight Moves                               |                            | Find the minimum knight moves from origin to a target on an infinite board.                                                                                              |
| 1202 | ★ | Smallest String With Swaps                         |                            | Swap characters at given index pairs to get the lexicographically smallest string.                                                                                       |
| 1245 | ★ | Tree Diameter                                      |                            | Find the diameter of an N-ary tree (longest path between any two nodes).                                                                                                 |
| 1293 | ★ | Shortest Path in a Grid with Obstacles Elimination |                            | Find the shortest path from start to end in a grid, eliminating at most k obstacles.                                                                                     |
| 1319 | ★ | Number of Operations to Make Network Connected     |                            | Find the minimum number of additional connections to make a network fully connected.                                                                                     |
| 1514 | ★ | #1514                                              | $O(E \log V) / O(V+E)$     | Max probability path between two nodes.                                                                                                                                  |
| 1559 | ★ | Detect Cycles in 2D Grid                           |                            | Detect if a 2D grid contains a cycle of the same character.                                                                                                              |
| 1631 | ★ | Path With Minimum Effort                           | $O(mn \log mn) / O(mn)$    | Find the path from top-left to bottom-right minimizing the maximum absolute difference.                                                                                  |

| #    | ★ | Name                                  | Complexity | Description                                                   |
|------|---|---------------------------------------|------------|---------------------------------------------------------------|
| 1743 | ★ | Restore the Array From Adjacent Pairs |            | Restore the original array from a list of all adjacent pairs. |

## Greedy (28 problems, 28 templated)

| #    | ★ | Name                                             | Complexity           | Description                                                                                                                                                 |
|------|---|--------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 45   | ★ | Jump Game II                                     | $O(n) / O(1)$        | Find the minimum number of jumps to reach the last index. Each element is max jump length.                                                                  |
| 55   | ★ | Jump Game                                        | $O(n) / O(1)$        | Given array of jump lengths, determine if you can reach the last index from index 0.                                                                        |
| 56   | ★ | Merge Intervals                                  | $O(n \log n) / O(n)$ | Merge all overlapping intervals in a collection and return non-overlapping sorted intervals.                                                                |
| 57   | ★ | Insert Interval                                  | $O(n) / O(n)$        | Insert a new interval into a sorted non-overlapping list of intervals, merging if necessary.                                                                |
| 134  | ★ | Gas Station                                      | $O(n) / O(1)$        | Find the starting gas station index from which you can complete a circular route. Return -1 if impossible.                                                  |
| 135  | ★ | Candy                                            |                      | Assign minimum candies to children in a line: each child 1 candy; child with higher rating than neighbor gets more.                                         |
| 252  | ★ | Meeting Rooms                                    | $O(n \log n) / O(1)$ | Determine if a person can attend all meetings (no overlapping intervals).                                                                                   |
| 253  | ★ | Meeting Rooms II                                 | $O(n \log n) / O(n)$ | Find the minimum number of conference rooms required to hold all meetings.                                                                                  |
| 274  | ★ | H-Index                                          |                      | Given citation counts, find the largest h such that h papers have citations each.                                                                           |
| 358  | ★ | Rearrange String k Distance Apart                | $O(n \log k) / O(k)$ | Rearrange a string so that the same characters are at least k distance apart. Return "" if impossible.                                                      |
| 406  | ★ | Queue Reconstruction by Height                   |                      | Reconstruct a queue of people given (height, k) pairs where k = taller-or-equal people in front.                                                            |
| 435  | ★ | Non-overlapping Intervals                        | $O(n \log n) / O(1)$ | Find the minimum number of intervals to remove so the rest don't overlap.                                                                                   |
| 452  | ★ | Minimum Number of Arrows to Burst Balloons       |                      | Find the minimum number of arrows to burst all balloons (each arrow goes through all balloons at that x).                                                   |
| 455  | ★ | Assign Cookies                                   |                      | Assign cookies greedily: each child needs minimum size, each cookie satisfies one child.                                                                    |
| 575  | ★ | Distribute Candies                               |                      | Distribute n candies (n/2 to eat) maximizing the number of distinct types.                                                                                  |
| 605  | ★ | Can Place Flowers                                |                      | Determine if n flowers can be planted in a flowerbed without adjacent flowers.                                                                              |
| 621  | ★ | Task Scheduler                                   | $O(n) / O(1)$        | Find the minimum intervals needed to execute n tasks with cooldown k between same-type tasks.                                                               |
| 630  | ★ | Course Schedule III                              |                      | Find the maximum courses you can take given (duration, deadline) pairs.                                                                                     |
| 646  | ★ | Maximum Length of Pair Chain                     |                      | Find the longest chain of pairs [a,b] where b < next_a.                                                                                                     |
| 763  | ★ | Partition Labels                                 |                      | Partition a string into as many parts as possible so each letter appears in at most one part.                                                               |
| 881  | ★ | Boats to Save People                             |                      | Find the minimum number of boats to rescue people given weight limit 2 per boat.                                                                            |
| 986  | ★ | Interval List Intersections                      |                      | Find all intersections of two lists of intervals.                                                                                                           |
| 1005 | ★ | Maximize Sum Of Array After K Negations          |                      | Given an array and k, perform exactly k operations, each negating one element (the same index may be chosen repeatedly), to maximize the array sum.         |
| 1094 | ★ | Car Pooling                                      |                      | Determine if a car pool trip is feasible given passenger pick-up and drop-off locations.                                                                    |
| 1272 | ★ | Remove Interval                                  |                      | Given a sorted list of disjoint intervals and an interval toBeRemoved, return the set of intervals after removing every point that lies inside toBeRemoved. |
| 1326 | ★ | Minimum Number of Taps to Open to Water a Garden |                      | Find the minimum number of taps to open to water the entire garden.                                                                                         |

| #    | ★ | Name                                          | Complexity | Description                                                     |
|------|---|-----------------------------------------------|------------|-----------------------------------------------------------------|
| 1353 | ★ | Maximum Number of Events That Can Be Attended |            | Find the maximum number of events you can attend (one per day). |
| 1488 | ★ | Avoid Flood in The City                       |            | Schedule drying days to prevent flooding given a list of rains. |

## Dynamic Programming (86 problems, 86 templated)

| #   | ★ | Name                                | Complexity        | Description                                                                                                                     |
|-----|---|-------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 5   | ★ | Longest Palindromic Substring       | $O(n^2) / O(1)$   | Find the longest palindromic substring within a given string.                                                                   |
| 10  | ★ | Regular Expression Matching         | $O(mn) / O(mn)$   | Implement regular expression matching with '.' (any single char) and '*' (zero or more of preceding). Must match entire string. |
| 44  | ★ | Wildcard Matching                   |                   | Match a string against a pattern with '?' (any single char) and '*' (any sequence including empty).                             |
| 53  | ★ | Maximum Subarray                    |                   | Find the contiguous subarray with the largest sum (at least one element).                                                       |
| 62  | ★ | Unique Paths                        | $O(mn) / O(n)$    | Count distinct paths from the top-left to the bottom-right of an $m \times n$ grid (only right or down moves).                  |
| 63  | ★ | Unique Paths II                     | $O(mn) / O(n)$    | Count distinct paths in an $m \times n$ grid with obstacles (0=open, 1=obstacle).                                               |
| 64  | ★ | Minimum Path Sum                    | $O(mn) / O(1)$    | Find the path from top-left to bottom-right of a grid with minimum sum.                                                         |
| 70  | ★ | Climbing Stairs                     | $O(n) / O(1)$     | Count the number of ways to climb $n$ stairs, taking 1 or 2 steps at a time.                                                    |
| 72  | ★ | Edit Distance                       | $O(mn) / O(mn)$   | Find the minimum number of edit operations (insert/delete/replace) to convert word1 to word2.                                   |
| 87  | ★ | #87                                 | $O(n^3) / O(n^2)$ | Is s2 a scramble of s1 (recursive splits)?                                                                                      |
| 91  | ★ | Decode Ways                         | $O(n) / O(1)$     | Count the number of ways to decode a string of digits into letters (A=1 ... Z=26).                                              |
| 96  | ★ | Unique Binary Search Trees          | $O(n^2) / O(n)$   | Count the number of structurally unique BSTs with exactly $n$ nodes.                                                            |
| 97  | ★ | Interleaving String                 |                   | Determine if s3 is formed by interleaving s1 and s2.                                                                            |
| 115 | ★ | Distinct Subsequences               | $O(mn) / O(mn)$   | Count distinct subsequences of s that equal t.                                                                                  |
| 120 | ★ | Triangle                            |                   | Find the minimum path sum from top to bottom of a triangle, moving to adjacent numbers on the row below.                        |
| 121 | ★ | Best Time to Buy and Sell Stock     | $O(n) / O(1)$     | Given daily stock prices, find the maximum profit from one buy-sell transaction (buy before sell).                              |
| 122 | ★ | Best Time to Buy and Sell Stock II  | $O(n) / O(1)$     | Maximize profit by completing as many buy-sell transactions as desired (no simultaneous holdings).                              |
| 123 | ★ | Best Time to Buy and Sell Stock III | $O(n) / O(1)$     | Maximize profit with at most two buy-sell transactions (no simultaneous holdings).                                              |
| 131 | ★ | Palindrome Partitioning             | $O(2^n) / O(n)$   | Partition a string into all possible subsets where every substring is a palindrome.                                             |
| 132 | ★ | Palindrome Partitioning II          |                   | Find the minimum number of cuts to partition a string so every part is a palindrome.                                            |
| 139 | ★ | Word Break                          | $O(n^2) / O(n)$   | Determine if a string can be segmented into space-separated words from a dictionary.                                            |
| 140 | ★ | Word Break II                       |                   | Return all ways to segment a string into space-separated words from a dictionary.                                               |
| 152 | ★ | Maximum Product Subarray            |                   | Find the contiguous subarray with the largest product.                                                                          |
| 174 | ★ | Dungeon Game                        |                   | Find the minimum initial health to traverse a dungeon grid from top-left to bottom-right.                                       |
| 188 | ★ | Best Time to Buy and Sell Stock IV  |                   | Maximize stock profit with at most $k$ buy-sell transactions.                                                                   |
| 198 | ★ | House Robber                        | $O(n) / O(1)$     | Maximize the amount stolen from houses in a row; cannot rob two adjacent houses.                                                |
| 213 | ★ | House Robber II                     | $O(n) / O(1)$     | Maximize robbery from a circular row of houses (first and last are adjacent).                                                   |

| #   | ★ | Name                                           | Complexity                              | Description                                                                                   |
|-----|---|------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------|
| 221 | ★ | Maximal Square                                 | $O(mn)$ / $O(mn)$                       | Find the largest square containing only 1s in a binary matrix. Return its area.               |
| 264 | ★ | Ugly Number II                                 |                                         | Find the nth ugly number (positive number whose prime factors are only 2, 3, 5).              |
| 279 | ★ | Perfect Squares                                | $O(n\sqrt{n})$ / $O(n)$                 | Find the minimum number of perfect squares summing to n.                                      |
| 300 | ★ | Longest Increasing Subsequence                 | $O(n^2)$ / $O(n)$                       | Find the length of the longest strictly increasing subsequence.                               |
| 309 | ★ | Best Time to Buy and Sell Stock with Cooldown  | $O(n)$ / $O(1)$                         | Find the maximum profit from stock with a 1-day cooldown after selling.                       |
| 312 | ★ | Burst Balloons                                 | $O(n^3)$ / $O(n^2)$                     | Burst all balloons to maximize collected coins. Coins = left * burst * right.                 |
| 313 | ★ | Super Ugly Number                              |                                         | Find the nth super ugly number (prime factors only from a given list).                        |
| 322 | ★ | Coin Change                                    | $O(\text{amount})$ / $O(\text{amount})$ | Find the minimum number of coins to make a given amount (coins can be reused).                |
| 337 | ★ | House Robber III                               |                                         | Find the maximum amount you can rob from a binary tree of houses (no two adjacent nodes).     |
| 343 | ★ | Integer Break                                  |                                         | Break n into at least two positive integers to maximize their product.                        |
| 368 | ★ | Largest Divisible Subset                       | $O(n^2)$ / $O(n)$                       | Find the largest subset where every pair has the larger divisible by the smaller.             |
| 375 | ★ | Guess Number Higher or Lower II                |                                         | Find the minimum cost to guarantee finding a number in range [1,n] using a guessing game.     |
| 376 | ★ | Wiggle Subsequence                             |                                         | Find the length of the longest wiggle subsequence (alternating up-down differences).          |
| 377 | ★ | Combination Sum IV                             | $O(\text{target})$ / $O(\text{target})$ | Count the number of possible combinations that add up to a target (order matters).            |
| 413 | ★ | Arithmetic Slices                              |                                         | Count the number of arithmetic slices (subarrays of length 3 with equal differences).         |
| 416 | ★ | Partition Equal Subset Sum                     | $O(n)$ / $O(\text{sum})$                | Determine if an integer array can be partitioned into two subsets with equal sum.             |
| 446 | ★ | Arithmetic Slices II - Subsequence             |                                         | Count the number of arithmetic slices that are subsequences of an array.                      |
| 474 | ★ | #474                                           | $O(l)$ / $O(mn)$                        | Max strings from list fitting within m zeros and n ones.                                      |
| 486 | ★ | Predict the Winner                             |                                         | Two players alternately pick from either end of an array; determine if the first player wins. |
| 494 | ★ | Target Sum                                     | $O(n)$ / $O(\text{sum})$                | Count the number of ways to assign + or - to array elements to reach a target sum.            |
| 509 | ★ | Fibonacci Number                               |                                         | Compute the nth Fibonacci number.                                                             |
| 516 | ★ | Longest Palindromic Subsequence                | $O(n^2)$ / $O(n^2)$                     | Find the length of the longest palindromic subsequence in a string.                           |
| 518 | ★ | Coin Change 2                                  | $O(\text{amount})$ / $O(\text{amount})$ | Count the number of combinations of coins that sum to an amount (coins may be reused).        |
| 552 | ★ | Student Attendance Record II                   |                                         | Count valid attendance records of length n (at most 1 absence, never 3+ consecutive lates).   |
| 576 | ★ | Out of Boundary Paths                          |                                         | Count paths from a given cell that leave an $m \times n$ grid in exactly maxMove steps.       |
| 583 | ★ | Delete Operation for Two Strings               | $O(mn)$ / $O(mn)$                       | Find the minimum number of deletions to make two strings equal.                               |
| 600 | ★ | Non-negative Integers without Consecutive Ones |                                         | Count integers in $[0, n]$ whose binary representation has no two adjacent 1 bits.            |
| 629 | ★ | K Inverse Pairs Array                          |                                         | Count permutations of 1..n with exactly k inverse pairs.                                      |
| 638 | ★ | Shopping Offers                                |                                         | Minimize the cost to purchase items given special-offer bundles.                              |

| #    | ★ | Name                                                      | Complexity             | Description                                                                                             |
|------|---|-----------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------|
| 639  | ★ | Decode Ways II                                            |                        | Count decodings of a digit string where * is a wildcard for digits 1-9, modulo 1e9+7.                   |
| 647  | ★ | Palindromic Substrings                                    | $O(n^2) / O(1)$        | Count all substrings of a string that are palindromes.                                                  |
| 650  | ★ | 2 Keys Keyboard                                           |                        | Find minimum steps to get exactly n 'A's on a screen using only Copy All and Paste.                     |
| 673  | ★ | Number of Longest Increasing Subsequence                  |                        | Find the number of longest increasing subsequences.                                                     |
| 674  | ★ | Longest Continuous Increasing Subsequence                 |                        | Find the length of the longest continuous (contiguous) strictly increasing subarray.                    |
| 688  | ★ | Knight Probability in Chessboard                          |                        | Find the probability that a knight stays on an $n \times n$ board after exactly k moves.                |
| 714  | ★ | Best Time to Buy and Sell Stock with Transaction Fee      | $O(n) / O(1)$          | Find the maximum profit from unlimited transactions with a transaction fee per trade.                   |
| 718  | ★ | Maximum Length of Repeated Subarray                       |                        | Find the maximum length of a subarray that appears in both arrays.                                      |
| 740  | ★ | Delete and Earn                                           |                        | Delete and earn: delete nums[i] to earn nums[i] points, but must delete all adjacent values.            |
| 746  | ★ | Min Cost Climbing Stairs                                  | $O(n) / O(1)$          | Find the minimum cost to climb a staircase (pay to step on, can skip one).                              |
| 787  | ★ | Cheapest Flights Within K Stops                           | $O(E \log E) / O(V+E)$ | Find the cheapest flight from src to dst with at most k stops.                                          |
| 873  | ★ | Length of Longest Fibonacci Subsequence                   |                        | Find the length of the longest Fibonacci-like subsequence in a sorted array.                            |
| 887  | ★ | Super Egg Drop                                            |                        | Find the minimum number of moves to determine the critical floor for egg drops.                         |
| 918  | ★ | Maximum Sum Circular Subarray                             |                        | Find the maximum subarray sum where the array is circular (subarray may wrap around).                   |
| 931  | ★ | #931                                                      | $O(mn) / O(1)$         | Min sum of a falling path through an $n \times n$ matrix.                                               |
| 935  | ★ | Knight Dialer                                             |                        | Count distinct phone numbers of length n that a knight can dial.                                        |
| 983  | ★ | Minimum Cost For Tickets                                  |                        | Find the minimum cost to cover all travel days using 1-, 7-, or 30-day passes.                          |
| 1027 | ★ | Longest Arithmetic Subsequence                            |                        | Find the length of the longest arithmetic subsequence in an array.                                      |
| 1048 | ★ | Longest String Chain                                      |                        | Find the longest string chain where each word is a predecessor of the next.                             |
| 1049 | ★ | #1049                                                     | $O(n) / O(\text{sum})$ | Smallest possible weight of last remaining stone.                                                       |
| 1137 | ★ | N-th Tribonacci Number                                    |                        | Return the nth Tribonacci number (each term is the sum of the previous three).                          |
| 1143 | ★ | Longest Common Subsequence                                | $O(mn) / O(mn)$        | Find the length of the longest common subsequence of two strings.                                       |
| 1216 | ★ | Valid Palindrome III                                      |                        | Check if a string is a k-palindrome (palindrome after removing at most k chars).                        |
| 1218 | ★ | Longest Arithmetic Subsequence of Given Difference        |                        | Find the longest arithmetic subsequence with a given difference.                                        |
| 1269 | ★ | Number of Ways to Stay in the Same Place After Some Steps |                        | Count the number of ways to stay at position 0 after exactly n steps on an array of length arrLen.      |
| 1312 | ★ | #1312                                                     | $O(n^2) / O(n^2)$      | Min insertions to make string a palindrome.                                                             |
| 1395 | ★ | Count Number of Teams                                     |                        | Count triplets (i, j, k) with $i < j < k$ whose ratings are strictly increasing or strictly decreasing. |

| #    | ★ | Name                                             | Complexity | Description                                                                 |
|------|---|--------------------------------------------------|------------|-----------------------------------------------------------------------------|
| 1567 | ★ | Maximum Length of Subarray With Positive Product |            | Find the maximum length subarray with a positive product.                   |
| 1696 | ★ | Jump Game VI                                     |            | Find the maximum score reaching the last index, jumping 1 to k steps.       |
| 1884 | ★ | Egg Drop With 2 Eggs and N Floors                |            | Find minimum moves to determine a critical floor using 2 eggs and n floors. |

## Trie (8 problems, 8 templated)

| #    | ★ | Name                                       | Complexity        | Description                                                                                                                                       |
|------|---|--------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 208  | ★ | Implement Trie (Prefix Tree)               | $O(L) / O(n)$     | Implement a Trie with insert, search, and startsWith operations.                                                                                  |
| 211  | ★ | Design Add and Search Words Data Structure | $O(L) / O(n)$     | Design a data structure supporting addWord and search (with '.' wildcard) for words.                                                              |
| 212  | ★ | Word Search II                             | $O(mn4^L) / O(L)$ | Find all words from a dictionary that exist in an $m \times n$ character board (adjacent cells, no reuse).                                        |
| 336  | ★ | Palindrome Pairs                           |                   | Find all palindrome pairs among a list of unique words.                                                                                           |
| 648  | ★ | Replace Words                              |                   | Given a sentence and a list of roots, replace each word with its root if one exists.                                                              |
| 676  | ★ | Implement Magic Dictionary                 |                   | Design a data structure with addWord and search supporting '.' wildcard.                                                                          |
| 720  | ★ | Longest Word in Dictionary                 |                   | Find the longest word in a dictionary that can be built one character at a time.                                                                  |
| 1233 | ★ | Remove Sub-Folders from the Filesystem     |                   | Given a list of folder paths, remove all sub-folders so only top-level folders remain. <code>"/a/b"</code> is a sub-folder of <code>"/a"</code> . |

## Bit Manipulation (16 problems, 16 templated)

| #   | ★ | Name                            | Complexity         | Description                                                                                                                                 |
|-----|---|---------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 136 | ★ | Single Number                   | $O(n) / O(1)$      | Find the single element in an array where every other element appears exactly twice.                                                        |
| 137 | ★ | Single Number II                |                    | Find the single element in an array where every other element appears exactly three times.                                                  |
| 190 | ★ | Reverse Bits                    |                    | Reverse all bits of a 32-bit unsigned integer.                                                                                              |
| 191 | ★ | Number of 1 Bits                | $O(1) / O(1)$      | Count the number of '1' bits in a 32-bit integer (Hamming weight).                                                                          |
| 201 | ★ | #201                            | $O(\log n) / O(1)$ | Bitwise AND of all numbers in range [left, right].                                                                                          |
| 231 | ★ | Power of Two                    | $O(1) / O(1)$      | Determine if a number is a power of two.                                                                                                    |
| 260 | ★ | Single Number III               | $O(n) / O(1)$      | Find two non-repeating numbers in an array where all others appear twice.                                                                   |
| 268 | ★ | Missing Number                  | $O(n) / O(1)$      | Find the single missing number in an array of n distinct integers in range [0, n].                                                          |
| 318 | ★ | Maximum Product of Word Lengths |                    | Find the maximum product of lengths of two words with no common letters.                                                                    |
| 338 | ★ | Counting Bits                   | $O(n) / O(n)$      | Return an array counting the number of 1 bits for each integer from 0 to n.                                                                 |
| 342 | ★ | Power of Four                   | $O(1) / O(1)$      | Determine if a number is a power of four.                                                                                                   |
| 371 | ★ | Sum of Two Integers             |                    | Add two integers without using + or - operators.                                                                                            |
| 405 | ★ | Convert a Number to Hexadecimal |                    | Given an integer <code>num</code> , return its hexadecimal representation as a lowercase string. For negative numbers use two's complement. |
| 461 | ★ | Hamming Distance                |                    | Calculate Hamming distance between two integers (positions where bits differ).                                                              |
| 476 | ★ | Number Complement               |                    | Find the complement of an integer (flip all bits up to the most significant bit).                                                           |
| 957 | ★ | Prison Cells After N Days       |                    | Find the prison cell configuration after n days of simultaneous updates.                                                                    |

## Design (25 problems, 25 templated)

| #    | ★ | Name                                                | Complexity                    | Description                                                                                                                                                                              |
|------|---|-----------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 146  | ★ | LRU Cache                                           | $O(1)$ / $O(\text{capacity})$ | Design an LRU (Least Recently Used) Cache with $O(1)$ get and put operations.                                                                                                            |
| 170  | ★ | Two Sum III - Data structure design                 |                               | Design a Two Sum data structure with <code>add()</code> and <code>find()</code> operations.                                                                                              |
| 173  | ★ | Binary Search Tree Iterator                         |                               | Implement a Binary Search Tree iterator with <code>hasNext()</code> and <code>next()</code> using $O(h)$ memory.                                                                         |
| 244  | ★ | Shortest Word Distance II                           |                               | Design data structure for finding shortest word distance between two words in a list, supporting repeated queries.                                                                       |
| 245  | ★ | Shortest Word Distance III                          |                               | Given a word list and two words (which may be equal), return the shortest index distance. When <code>word1 == word2</code> , find the closest pair of distinct occurrences of that word. |
| 251  | ★ | Flatten 2D Vector                                   |                               | Design an iterator over a 2D vector with <code>next()</code> and <code>hasNext()</code> that yields all elements in row-major order.                                                     |
| 281  | ★ | Zigzag Iterator                                     |                               | Given two lists, design an iterator that returns elements alternating between them; when one is exhausted, continue with the other.                                                      |
| 341  | ★ | Flatten Nested List Iterator                        |                               | Implement an iterator that flattens a nested list of integers.                                                                                                                           |
| 346  | ★ | Moving Average from Data Stream                     |                               | Compute the moving average from a stream of integers given a window size.                                                                                                                |
| 348  | ★ | Design Tic-Tac-Toe                                  |                               | Design a Tic-Tac-Toe game that detects a winner in $O(1)$ per move.                                                                                                                      |
| 380  | ★ | Insert Delete GetRandom $O(1)$                      |                               | Design a set with $O(1)$ insert, remove, and random element retrieval.                                                                                                                   |
| 381  | ★ | Insert Delete GetRandom $O(1)$ - Duplicates allowed |                               | Same as 380 but allows duplicates.                                                                                                                                                       |
| 460  | ★ | LFU Cache                                           | $O(1)$ / $O(\text{capacity})$ | Design an LFU (Least Frequently Used) Cache with $O(1)$ get and put.                                                                                                                     |
| 519  | ★ | Random Flip Matrix                                  |                               | Design a matrix with random flip and reset; each 0-cell can be flipped to 1 exactly once.                                                                                                |
| 622  | ★ | Design Circular Queue                               |                               | Design a circular queue (ring buffer) with fixed capacity.                                                                                                                               |
| 641  | ★ | Design Circular Deque                               |                               | Design a double-ended circular queue with <code>maxSize</code> capacity.                                                                                                                 |
| 705  | ★ | Design HashSet                                      |                               | Design a HashSet with add, remove, and contains operations.                                                                                                                              |
| 706  | ★ | Design HashMap                                      |                               | Design a HashMap with put, get, and remove operations.                                                                                                                                   |
| 707  | ★ | Design Linked List                                  |                               | Design a doubly linked list with index-based get, <code>addAtHead</code> , <code>addAtTail</code> , <code>addAtIndex</code> , <code>deleteAtIndex</code> .                               |
| 729  | ★ | 我的日程安排表 I                                           |                               | Book time intervals on a calendar; return false if overlap exists.                                                                                                                       |
| 731  | ★ | 我的日程安排表 II                                          |                               | Book events; return number of existing bookings overlapping the new booking.                                                                                                             |
| 732  | ★ | My Calendar III                                     |                               | Book events; return the maximum k-booking (k events overlap at same time).                                                                                                               |
| 745  | ★ | #745                                                | $O(L)$ / $O(n)$               | Prefix and suffix search --- find word with given prefix and suffix.                                                                                                                     |
| 933  | ★ | Number of Recent Calls                              |                               | Find the number of requests made in the last 3000 milliseconds.                                                                                                                          |
| 1570 | ★ | Dot Product of Two Sparse Vectors                   |                               | Compute the dot product of two sparse vectors efficiently.                                                                                                                               |

## Math (57 problems, 57 templated)

| #   | ★ | Name                             | Complexity                | Description                                                                                                                                                                   |
|-----|---|----------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9   | ★ | Palindrome Number                | $O(\log n) / O(1)$        | Determine whether an integer reads the same forward and backward. Negative numbers are never palindromes.                                                                     |
| 29  | ★ | Divide Two Integers              |                           | Divide two integers without using multiplication, division, or mod. Clamp to 32-bit signed range.                                                                             |
| 50  | ★ | Pow(x, n)                        | $O(\log n) / O(1)$        | Implement $\text{pow}(x, n)$ --- raise $x$ to the power $n$ ( $n$ can be negative).                                                                                           |
| 59  | ★ | Spiral Matrix II                 |                           | Given $n$ , generate an $n \times n$ matrix filled with the numbers $1$ to $n \times n$ in spiral order.                                                                      |
| 66  | ★ | Plus One                         |                           | Increment a large integer represented as an array of digits.                                                                                                                  |
| 118 | ★ | Pascal's Triangle                |                           | Generate Pascal's triangle up to <code>numRows</code> rows.                                                                                                                   |
| 119 | ★ | Pascal's Triangle II             |                           | Return the $k$ th row of Pascal's triangle using $O(k)$ space.                                                                                                                |
| 149 | ★ | Max Points on a Line             |                           | Find the maximum number of points that lie on the same straight line.                                                                                                         |
| 172 | ★ | Factorial Trailing Zeroes        |                           | Count trailing zeroes in $n!$ (count factors of 5).                                                                                                                           |
| 202 | ★ | Happy Number                     |                           | Determine if a number eventually reaches 1 by repeatedly replacing it with the sum of squares of its digits.                                                                  |
| 204 | ★ | Count Primes                     | $O(n \log \log n) / O(n)$ | Count prime numbers less than a non-negative integer $n$ .                                                                                                                    |
| 223 | ★ | Rectangle Area                   |                           | Given the coordinates of two axis-aligned rectangles, return the total area they cover (counting the overlap once).                                                           |
| 233 | ★ | Number of Digit One              |                           | Count total occurrences of digit 1 in all integers from 1 to $n$ .                                                                                                            |
| 243 | ★ | Shortest Word Distance           |                           | Given an array of words and two distinct words, return the shortest distance (in indices) between any occurrence of the two words.                                            |
| 258 | ★ | Add Digits                       |                           | Repeatedly sum digits until a single digit remains (digital root).                                                                                                            |
| 263 | ★ | Ugly Number                      |                           | Check if a number is an ugly number (prime factors only 2, 3, 5).                                                                                                             |
| 292 | ★ | Nim Game                         |                           | In Nim game, you and opponent alternate removing 1--3 stones; player who takes last stone wins. Can you win?                                                                  |
| 319 | ★ | Bulb Switcher                    |                           | $n$ bulbs toggled by $n$ rounds (round $i$ toggles every $i$ th bulb). Count bulbs on after all rounds.                                                                       |
| 326 | ★ | Power of Three                   |                           | Check if a number is a power of three.                                                                                                                                        |
| 357 | ★ | Count Numbers with Unique Digits |                           | Given $n$ , count all numbers $x$ with $0 \leq x < 10^n$ that have no repeated digits.                                                                                        |
| 365 | ★ | Water and Jug Problem            |                           | Given two jugs of capacities $x$ and $y$ and a target <code>target</code> , determine if you can measure exactly <code>target</code> liters using fill/empty/pour operations. |
| 367 | ★ | Valid Perfect Square             |                           | Given a positive integer <code>num</code> , return whether it is a perfect square, without using any built-in <code>sqrt</code> function.                                     |
| 372 | ★ | Super Pow                        | $O(\log n) / O(1)$        | Compute $a^b \bmod 1337$ where $b$ is given as an array of digits (so $b$ may be astronomically large).                                                                       |
| 397 | ★ | Integer Replacement              |                           | Find the minimum steps to reduce a number to 1 (even: $n/2$ , odd: $n+1$ or $n-1$ ).                                                                                          |
| 398 | ★ | Random Pick Index                |                           | Pick a random index from those matching a target value (unknown total size).                                                                                                  |
| 400 | ★ | Nth Digit                        |                           | Given $n$ , return the $n$ -th digit in the infinite sequence of concatenated positive integers <code>123456789101112...</code> .                                             |
| 412 | ★ | Fizz Buzz                        |                           | Print numbers 1-- $n$ , replacing multiples of 3 with 'Fizz', 5 with 'Buzz', both with 'FizzBuzz'.                                                                            |

| #    | ★ | Name                                        | Complexity             | Description                                                                                                                                                                                                        |
|------|---|---------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 441  | ★ | Arranging Coins                             |                        | You have <code>n</code> coins to form a staircase where the <code>k</code> -th row has exactly <code>k</code> coins; return the number of complete rows.                                                           |
| 453  | ★ | Minimum Moves to Equal Array Elements       |                        | In one move you increment <code>n-1</code> elements of the array by 1; return the minimum moves to make all elements equal.                                                                                        |
| 458  | ★ | Poor Pigs                                   |                        | With <code>buckets</code> buckets one of which is poisoned, <code>minutesToDie</code> for poison to act, and <code>minutesToTest</code> total time, return the minimum pigs needed to find the poisoned bucket.    |
| 470  | ★ | Implement Rand10() Using Rand7()            |                        | Implement rand10() using only rand7().                                                                                                                                                                             |
| 492  | ★ | Construct the Rectangle                     |                        | Given a target <code>area</code> , return the dimensions <code>[L, W]</code> with <code>L &gt;= W</code> , <code>L * W == area</code> , and <code>L - W</code> minimized.                                          |
| 495  | ★ | Teemo Attacking                             |                        | Given sorted attack <code>timeSeries</code> and a poison <code>duration</code> , return the total time the target is poisoned (overlaps counted once).                                                             |
| 528  | ★ | Random Pick with Weight                     |                        | Randomly pick an index proportional to its weight.                                                                                                                                                                 |
| 593  | ★ | Valid Square                                |                        | Given four points, return whether they form a valid square (positive area).                                                                                                                                        |
| 598  | ★ | Range Addition II                           |                        | Count cells in the intersection of all rectangles defined by range addition operations.                                                                                                                            |
| 633  | ★ | Sum of Square Numbers                       |                        | Given a non-negative integer <code>c</code> , decide whether there exist non-negative integers <code>a</code> , <code>b</code> with <code>a*a + b*b == c</code> .                                                  |
| 656  | ★ | Coin Path                                   |                        | Given <code>coins</code> (cost per index, <code>-1</code> blocked) and a max jump <code>maxJump</code> , find the lexicographically smallest cheapest path of indices from index 0 to the last.                    |
| 781  | ★ | Rabbits in Forest                           |                        | Each surveyed rabbit reports how many other rabbits share its color; given the <code>answers</code> , return the minimum number of rabbits in the forest.                                                          |
| 789  | ★ | Escape The Ghosts                           |                        | Starting at the origin with a <code>target</code> , and given <code>ghosts</code> positions, you all move simultaneously in Manhattan steps; return whether you can reach the target before any ghost catches you. |
| 792  | ★ | Number of Matching Subsequences             |                        | Given a string <code>s</code> and an array of <code>words</code> , return how many words are subsequences of <code>s</code> .                                                                                      |
| 829  | ★ | Consecutive Numbers Sum                     |                        | Find the number of ways to write <code>n</code> as a sum of consecutive positive integers.                                                                                                                         |
| 836  | ★ | Rectangle Overlap                           |                        | Check if two axis-aligned rectangles overlap.                                                                                                                                                                      |
| 858  | ★ | Mirror Reflection                           |                        | Given clock hands at <code>h:m</code> , find the minimum angle between them.                                                                                                                                       |
| 869  | ★ | Reordered Power of 2                        |                        | Check if any permutation of <code>n</code> 's digits is a power of 2.                                                                                                                                              |
| 939  | ★ | Minimum Area Rectangle                      |                        | Find the minimum area rectangle from a set of points in an axis-aligned grid.                                                                                                                                      |
| 963  | ★ | Minimum Area Rectangle II                   |                        | Find the minimum area rectangle aligned to any angle from a set of points.                                                                                                                                         |
| 1041 | ★ | Robot Bounded In Circle                     |                        | Determine if a robot following instructions stays within a bounded circle.                                                                                                                                         |
| 1160 | ★ | Find Words That Can Be Formed by Characters |                        | Given <code>words</code> and a string <code>chars</code> , return the total length of words that can be formed using letters of <code>chars</code> (each letter used at most as often as it appears).              |
| 1201 | ★ | Ugly Number III                             | $O(\log(\min)) / O(1)$ | Find the <code>nth</code> positive integer that is divisible by at least one of <code>a</code> , <code>b</code> , or <code>c</code> .                                                                              |
| 1304 | ★ | Find N Unique Integers Sum up to Zero       |                        | Find <code>n</code> unique integers that sum to zero.                                                                                                                                                              |
| 1344 | ★ | Angle Between Hands of a Clock              |                        | Calculate the angle between clock hands at a given time.                                                                                                                                                           |
| 1583 | ★ | Count Unhappy Friends                       |                        | Count the number of unhappy friends after pairing.                                                                                                                                                                 |
| 1588 | ★ | Sum of All Odd Length Subarrays             |                        | Compute the sum of all odd-length subarray sums.                                                                                                                                                                   |

| #    | ★ | Name                                           | Complexity | Description                                                                                                                              |
|------|---|------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1646 | ★ | Get Maximum in Generated Array                 |            | Generate an array from a given sequence and return the max element.                                                                      |
| 1823 | ★ | Find the Winner of the Circular Game           |            | <code>n</code> friends in a circle eliminate every <code>k</code> -th friend repeatedly; return the 1-indexed winner (Josephus problem). |
| 1969 | ★ | Minimum Non-Zero Product of the Array Elements |            | Find the minimum non-zero product of array elements after swaps.                                                                         |

Concurrency (no dedicated template) (2 problems, 0 templated)

| #    | ★ | Name                     | Complexity | Description |
|------|---|--------------------------|------------|-------------|
| 1114 |   | Print in Order           |            |             |
| 1115 |   | Print FooBar Alternately |            |             |

Total: 737 problems indexed across 15 categories.